



An interpretable deep learning model for EEG emotion recognition using the Rashomon effect

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ABSTRACT

Traditional EEG emotion recognition pipelines typically focus on identifying a single best-performing model, implicitly assuming that such a model provides unique scientific insight. This work embraces this perspective by constructing Rashomon set ($\mathcal{R}$) of high-performing deep graph neural networks (GNNs) for EEG-based emotion recognition across two benchmark datasets: DEAP (32 channels, 32 subjects) and DREAMER (14 channels, 23 subjects). Beyond classification, we quantify interpretability diversity using SHAP-based node-lobe attributions and introduce the Pairwise Dissimilarity Index (PDI) to measure explanation-level divergence across Rashomon set ($\mathcal{R}$).

The proposed **Rashomon-GNN** achieves state-of-the-art LOSO performance on the DEAP dataset: 90.52%, 91.96%, and 92.20% F1 for arousal, valence, and dominance respectively. Rashomon-GNN achieves absolute F1 improvements of +0.90% on arousal and +1.94% on valence, while requiring nearly an order of magnitude fewer parameters (5926 vs. 74,400). Against shallower baselines (TSception, EEGNet), improvements exceed 27 percentage points in F1, confirming consistent superiority across a diverse set of competing methods. On DREAMER, the framework achieves 59.8%, 58.1%, and 60.6% F1, demonstrating generalization despite the substantially reduced channel density (14 vs. 32 electrodes). Cross-dataset analysis reveals that while DEAP benefits from richer spatial resolution, both datasets exhibit substantial Rashomon diversity (17%–31% of pipelines within top-5% performance on DEAP; 17%–23% on DREAMER), confirming that model multiplicity is a dataset-agnostic structural property of the combinatorial GNN pipeline space.

SHAP and PDI analyses reveal that top-performing models converge on distinct but neuroscientifically meaningful EEG regions, whereas low-performing models exhibit redundant or diffuse attributions. Rashomon set diversity remains stable across $\epsilon \in \{0.01, 0.10\}$. Overall, this study demonstrates that multiple, computationally efficient GNN pipelines can achieve high accuracy while offering complementary and interpretable neurophysiological insights which opens new directions for trustworthy, robust, and explanation-aware affective computing.

1. Introduction

Emotion recognition from electroencephalography (EEG) signals plays a critical role in affective computing, human–computer interaction, and neuropsychiatric assessment. EEG captures non-invasive recordings of brain activity, offering rich temporal information about cognitive and affective states. However, interpreting these signals to detect emotional states remains a grand challenge due to the high dimensionality, non-stationarity, and subject-dependence of EEG data [1, 2]. These issues also explain why multiple representations of emotional states may coexist across subjects and time, making EEG a fertile ground for exploring model multiplicity.

Traditional pipelines for EEG-based emotion recognition have predominantly focused on identifying a single best model based on performance metrics like accuracy or F1. This strategy, however, may ignore the broader landscape of equally viable solutions and thereby restrict interpretability, generalizability, and scientific insight [3–5]. Recent deep computational predictors such as DeepAIPs-SFLA, PACPS-DNN, TargetCLP, SBSM-Pro, pACP-HybDeep, and PNPS-CAPSNET have demonstrated strong performance in biomedical signal classification, but continue the paradigm of pursuing a single optimal model [6–11]. This gap motivates a shift toward evaluating *sets* of high-performing models rather than one dominant solution.

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Recent advancements in machine learning theory have emphasized the concept of the *Rashomon effect*, a phenomenon where many models can achieve nearly identical predictive performance while relying on fundamentally different internal logic [12,13]. A *Rashomon set* ($\mathcal{R}$) consists of all models lying within a small performance margin (e.g., top 5%) of the best model [14,15]. For EEG-based emotion recognition, this concept is particularly relevant: emotional processing may involve multiple neural pathways, and different models may emphasize distinct spectral bands, spatial topologies, or feature modalities while achieving similar accuracy. Despite this, no existing EEG emotion recognition work has systematically explored Rashomon set ($\mathcal{R}$) or examined how diverse high-performing models differ in their internal decision logic [16].

A broad spectrum of feature extraction methods has been used in EEG emotion analysis, including bandpower features, differential entropy, fractal dimensions, Hjorth parameters, and wavelet/TQWT-based decompositions [17–19]. These handcrafted methods provide diverse spectral and temporal perspectives, yet the interplay between feature diversity and model diversity remains underexplored. Moreover, existing studies rarely investigate whether particular feature-graph combinations lead to functionally distinct but equally accurate models.

Graph Neural Networks (GNNs) have recently emerged as a powerful architecture for modeling EEG as brain graphs, where nodes represent electrodes (or brain regions) and edges encode inter-channel functional connectivity [20,21]. Unlike conventional deep learning approaches, GNNs naturally integrate spatial dependencies and support structured representation learning. By combining GNNs with modular pipelines for feature and graph construction, researchers can in principle explore a vast combinatorial hypothesis space, although prior work has focused primarily on demonstrating performance improvements rather than understanding model diversity [22,23].

Explainability techniques such as SHAP have become instrumental in understanding how different models make decisions, especially in biomedical domains where interpretability is as important as accuracy. SHAP offers a principled approach to attributing output predictions to input features, enabling researchers to visualize and compare model rationales across different model configurations [24–26]. SHAP has been successfully applied to EEG-based emotion recognition [27, 28], yet no study has assessed SHAP-driven interpretability within a Rashomon framework or quantified how explanation structures vary across high-performing graph-based models.

Several recent studies have demonstrated the benefits of exploring multiple modeling pathways in EEG tasks. Ensemble learning frameworks have improved robustness across subjects [4,29], and GNN-based models have outperformed traditional CNN or RNN-based architectures in capturing spatial and functional brain dynamics [20,23]. However, existing studies treat these models independently and do not analyze the diversity and interpretability of the model space itself an aspect that is critical for neuroscientific insight, personalization, and trustworthy deployment.

In this paper, we present a systematic Rashomon analysis of deep graph learning models for EEG-based emotion recognition. We construct a broad set of models by pairing multiple node-level feature extraction methods, such as Lorentzian bandpower, Hjorth parameters, entropy measures, and wavelet-based features, with different graph construction strategies, including PLV, coherence, correlation, mutual information, and amplitude envelope correlation. Each feature-graph pairing defines a distinct pipeline, which is then processed through a graph convolutional network equipped with layer-wise fusion and a multi-head output layer.

Our framework evaluates predictive performance under leave-one-subject-out (LOSO) validation and applies SHAP (SHapley Additive exPlanations) to assess model interpretability and quantify diversity within the Rashomon set ($\mathcal{R}$) [24,27,28]. We demonstrate that members of the Rashomon set ($\mathcal{R}$) vary substantially in feature reliance and

connectivity structure, reinforcing the view that emotional states may manifest through multiple, non-exclusive neural mechanisms.

By embracing the Rashomon effect, our work advocates a paradigm shift in EEG emotion recognition from singular model pursuit to systematic model diversity and interpretability. This perspective aligns with emerging trends in trustworthy AI, biomedical reproducibility, and neuroscience-informed machine learning.

Contributions

The main contributions and novelties of this work are summarized as follows. The core novelty is threefold: (1) we construct and analyze a *Rashomon set* over a combinatorial GNN pipeline space for EEG emotion recognition; (2) we introduce the Pairwise Dissimilarity Index (PDI) as a principled measure of attribution diversity within high-performing model sets; and (3) we validate both phenomena across two benchmark datasets with rigorous LOSO evaluation, SHAP-driven interpretability, and comprehensive computational analysis. Detailed contributions are:

- a. Rashomon-based GNN framework for EEG emotion recognition.** We exhaustively explore 48 feature-graph pipelines across DEAP and DREAMER under LOSO, demonstrating that multiple high-performing models co-exist rather than a single optimum.
- b. Pairwise Dissimilarity Index (PDI).** A new metric quantifying explanation-level divergence among Rashomon-set members. Top-5 models exhibit 100× higher PDI than bottom-5, confirming that high accuracy and functional diversity co-occur.
- c. Hybrid SHAP/GNNExplainer interpretability module.** A unified post-hoc explainability layer provides both feature-level and graph-structural attributions. Across 7920 SHAP runs (4608 DEAP + 3312 DREAMER), target-specific lobe-level activation patterns are consistently identified.
- d. State-of-the-art LOSO performance with minimal parameters.** Rashomon-GNN achieves F1 of 90.52%, 91.96%, 92.20% (DEAP) and 59.8%, 58.1%, 60.6% (DREAMER) using only 5.9K parameters—12× fewer than Diff-MT with no architecture changes between datasets.
- e. Comprehensive computational and ablation analysis.** $O(n)$ scalability, 15 ms inference latency, and 6.3 GB peak memory are reported alongside fusion-layer and ϵ -threshold ablations (all three fusion strategies achieve ~54.2% LOSO accuracy, $\Delta < 2\%$) and cross-dataset statistical comparisons (Cohen's $d > 2.5$).

2. Related work

2.1. EEG emotion recognition with GNNs

Graph neural networks (GNNs) have become a dominant paradigm in EEG-based emotion recognition due to their ability to represent inter-channel dependencies as structured brain connectivity graphs [22,30–33].

Song et al. [22] introduced the Dynamic Graph Convolutional Neural Network (DGCNN), which adaptively learns adjacency matrices during training, capturing subject-specific spatial and temporal dynamics. Zhong et al. [30] incorporated anatomical priors to regularize edge structures, enhancing physiological plausibility.

Li et al. [31] proposed Self-Organizing GNNs (SOGNNs), enabling dynamic adjustment of node features and graph topology for personalized emotion modeling. Chen et al. [32] introduced dual channel-frequency attention mechanisms to capture both spectral salience and topological relevance. These studies highlight the growing sophistication of GNNs in modeling both node-level features and connectivity structures.

Mathematically, a GNN processes a graph $G = (V, E, X)$, where $X \in \mathbb{R}^{N \times d}$ denotes the feature matrix for N EEG nodes and E represents

functional connectivity (e.g., PLV, MI); the complete graph convolution update is defined formally in Eq. (18).

While Liu et al. [33] provide an extensive survey of GNN variants for EEG emotion recognition, prior work largely emphasizes graph construction or temporal modeling. Crucially, these studies do not examine how combining node-level feature extractors with graph construction methods can produce multiple functionally distinct yet similarly performing models. This is a key question addressed in this paper.

Wavelet and TQWT-based EEG emotion recognition. Alongside GNN-based methods, traditional emotion recognition research relies on filter-bank energy, wavelet packet decomposition, and tunable Q-factor wavelet transform (TQWT) subbands. Wavelet-based pipelines extract multi-resolution oscillatory features such as wavelet energy, entropy, and subband statistics, showing strong performance across DEAP and DREAMER datasets [17,18]. TQWT-based decompositions further enable tunable oscillatory representations suited for emotional state discrimination [19]. However, these methods do not leverage graph structure, and more importantly, they do not investigate the combinatorial diversity arising from pairing multiple feature extractors with multiple graph-building strategies a central novelty of our work. In contrast, Rashomon-GNN explicitly maps wavelet-derived features (e.g., Daubechies wavelet energy and subband entropy) as node-level attributes on EEG functional connectivity graphs, unifying spectral richness with graph topology. By exhaustively pairing these wavelet feature families with six connectivity estimators (PLV, coherence, correlation, MI, PLI, AEC) under LOSO validation, our framework identifies which spectral-topological combinations populate the Rashomon set an analysis entirely absent from prior wavelet and TQWT-only pipelines, which optimize a single model rather than characterizing the full high-performing model space.

Deep predictors in biomedical and affective modeling. Recent deep computational predictors such as DeepAIPs-SFLA, PACPS-DNN, TargetCLP, SBSM-Pro, pACP-HybDeep, and PNPS-CAPSNET have advanced sequence and biomedical signal classification [6–11]. While powerful, these approaches focus on building a single optimal model and do not explore explanation diversity, Rashomon set ($\mathcal{R}$), or multi-graph/multi-feature interactions. Incorporating these works strengthens the context for deep-learning-based model diversity analysis.

2.2. Explainable AI in EEG using SHAP

Explainability is essential for neuroscientific trust and clinical deployment. SHAP (SHapley Additive exPlanations) [24] provides feature attribution scores ϕ_i grounded in cooperative game theory, satisfying the axioms of local accuracy, missingness, and consistency; the formal Shapley definition and implementation details are given in the Methods section.

Sylvester et al. [27] used SHAP to identify temporal ERP components. Miao et al. [28] developed ST-SHAP to explain hierarchical spatio-temporal Transformers, while Dokare and Gupta [34] demonstrated SHAP's applicability across clinical EEG tasks.

Although SHAP has been widely applied to conventional deep models, its integration with GNN-based EEG pipelines remains limited. Existing XAI studies primarily analyze a single model at a time and do not investigate explanation-level diversity across multiple high-performing pipelines one of the core motivations of our Rashomon-GNN analysis.

2.3. Rashomon set ($\mathcal{R}$) and interpretability

The Rashomon effect indicates that multiple distinct models can achieve comparable predictive accuracy [12,13], thereby challenging the assumption that a high-performing model offers a unique scientific explanation.

Semenova et al. [14] formalized the *Rashomon ratio*, defined as

$$R(\epsilon) = \frac{|\{m_k \in \mathcal{M} : \mathcal{L}_k \leq (1 + \epsilon)\mathcal{L}^*\}|}{|\mathcal{M}|}, \quad (1)$$

where $\mathcal{L}^*$ is the minimal loss within the model class $\mathcal{M}$.

Laberge et al. [15] introduced robust feature importance rankings derived from ensembles of top-performing models. Donnelly et al. [35] demonstrated the fairness and transparency advantages of Rashomon-set surfacing.

However, Rashomon analysis has received very little attention in EEG-based emotion recognition and, to the best of our knowledge, has not been explored at all in graph-based EEG modeling. In this work, we try to close that gap by simply putting different feature sets together with different graph constructions and seeing which combinations perform similarly well. From there, we identify the Rashomon set ($\mathcal{R}$) under Leave-One-Subject-Out (LOSO) validation and then look more closely at how these models differ in terms of brain-region importance, using SHAP explanations and subject-level dissimilarity analysis.

3. Methods

3.1. Datasets and preprocessing

DEAP dataset. The DEAP (Dataset for Emotion Analysis using EEG, Physiological and Video Signals) dataset contains 32 participants watching 40 one-minute music video clips. Each participant provided self-assessment ratings for valence, arousal, dominance, and liking on a scale from 1 to 9. EEG signals were recorded from 32 channels at 512 Hz and preprocessed to 128 Hz. We use the preprocessed version with 32 subjects and 40 trials per subject, totaling 1280 trials.

DREAMER dataset. The DREAMER (Database for Emotion Recognition through EEG and ECG signals from Wireless and Low-cost devices) dataset comprises 23 participants watching 18 emotional film clips. Each clip elicits specific emotional responses across the valence-arousal-dominance space. EEG signals were recorded from 14 channels (Fp1, Fp2, F3, F4, F7, F8, C3, C4, T7, T8, P3, P4, O1, O2) at 128 Hz. The dataset contains baseline (61 s) and stimuli (clips varying 65–393 s). We use all 23 subjects with 18 trials per subject, totaling 414 trials.

Preprocessing and label construction. To preserve richer oscillatory dynamics and connectivity structure, all EEG trials are standardized to $T = 256$ samples per channel (after the same fixed-duration segmenting used in our protocol) prior to feature extraction and graph construction. At the same time, it stays aligned with commonly followed evaluation protocols, making our results easier to reproduce. The downsampling and standardization step strikes a practical balance: it reduces computational cost without discarding the key characteristics of the EEG signals. This kind of preprocessing is widely used in EEG pipelines [36].

We follow the Valence-Arousal-Dominance (VAD) formulation and binarize the ground-truth labels by thresholding each axis into high/low bins for multi-label classification, as illustrated in Fig. 1. Each trial is thus associated with three binary labels corresponding to Valence, Arousal, and Dominance.

Cross-dataset generalization scope. To rigorously assess dataset dependence and generalization, all experiments are performed independently and completely on both DEAP and DREAMER. A joint statistical comparison is provided in Section 5.6, which explicitly examines how differences in channel density (32 vs. 14), stimulus modality (audio-visual vs. film clips), and subject count (32 vs. 23) affect predictive performance, Rashomon set size, and PDI diversity.

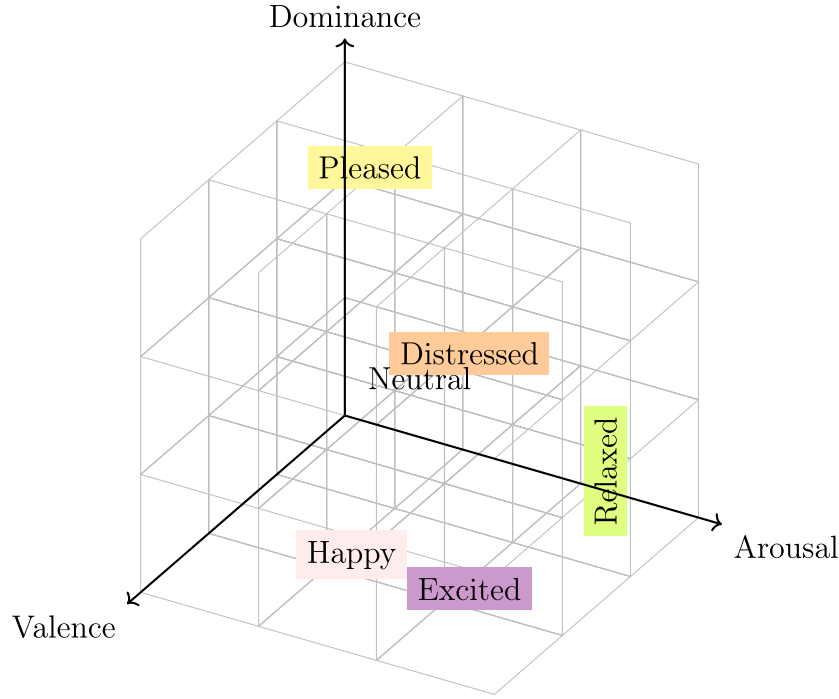


Fig. 1. 3D affective space representing six emotions derived from the Valence-Arousal-Dominance (VAD) model. Emotion labels are obtained by thresholding each axis into high and low bins. Used for binarizing DEAP ground truth labels for classification.

3.2. Feature extraction

For each EEG channel (node), we compute a set of complementary node-level features that capture oscillatory energy, spectral complexity, and time-domain dynamics. These features are motivated by prior evidence in affective EEG: relative band-power has been shown to discriminate affective states [37], differential entropy is frequently competitive on DEAP/SEED [38], wavelet-based entropy and fractal descriptors capture nonlinear complexity [39], and Hjorth parameters have a long history in real-time EEG analysis [40].

Hjorth mobility. Given a channel signal $x(t)$, Hjorth mobility is computed as:

$$\text{Mobility}(x) = \sqrt{\frac{\text{var}(\dot{x}(t))}{\text{var}(x(t))}}, \quad (2)$$

where $\dot{x}(t)$ denotes the first temporal derivative and $\text{var}(\cdot)$ is variance.

Spectral entropy. Let p_i be the normalized power in frequency bin i (i.e., $\sum_i p_i = 1$). Spectral entropy is:

$$H(x) = - \sum_i p_i \log(p_i), \quad (3)$$

where the log base can be natural unless otherwise specified.

3.3. Graph construction

For each trial, we construct multiple functional connectivity graphs among EEG channels, including phase locking value (PLV), Pearson correlation, spectral coherence, mutual information (MI), phase lag index (PLI), and amplitude envelope correlation (AEC). These connectivity measures reflect distinct neurophysiological hypotheses regarding emotion-related brain interactions [33]. Each connectivity matrix is treated as a weighted adjacency matrix; when required, we symmetrize it and add self-loops before GCN normalization. We retain weighted edges (no hard thresholding) and apply standard GCN normalization using $\hat{A} = A + I$ and its degree matrix $\hat{D}$. Each connectivity definition g_j is evaluated as a separate adjacency matrix A within the (f_i, g_j)

configuration space; we do not merge graph types unless explicitly stated.

Phase locking value (PLV). Let $\phi_i(t)$ and $\phi_j(t)$ be the instantaneous phases of channels i and j at time t (e.g., via the analytic signal). PLV is defined as:

$$\text{PLV}_{ij} = \left| \frac{1}{T} \sum_{t=1}^T \exp(j(\phi_i(t) - \phi_j(t))) \right|, \quad (4)$$

where j is the imaginary unit and T is the number of time samples used to estimate PLV.

Consistent with evidence that spatial proximity does not necessarily reflect functional coupling, dynamic or learned graph constructions have also been advocated in the literature [33]. In contrast to learning a single graph topology (e.g., DGCNN [22]), our framework explicitly explores multiple connectivity definitions to test diverse topological hypotheses.

3.4. Rashomon set ($\mathcal{R}$) construction

Let $\mathcal{F} = \{f_1, \dots, f_m\}$ denote the set of node feature extractors and $\mathcal{G} = \{g_1, \dots, g_n\}$ the set of graph construction methods. Each pair (f_i, g_j) yields a distinct configuration, producing graphs G_{ij} for all trials. For each configuration, we train and evaluate a model under leave-one-subject-out (LOSO), store its predictions, metrics, and trained parameters, and then identify the Rashomon set ($\mathcal{R}$).

Let $F1^{(f_i, g_j)}$ denote the mean LOSO F1 (aggregated across held-out subjects) for configuration (f_i, g_j) , and let

$$F^* = \max_{i,j} F1^{(f_i, g_j)}. \quad (5)$$

We define the Rashomon set ($\mathcal{R}$) as $\mathcal{R} = \{(f_i, g_j) \in \mathcal{F} \times \mathcal{G} : F1^{(f_i, g_j)} \geq (1 - \epsilon)F^*\}$ (see Eq. (23) for the full formal statement).

In our experiments, we use $\epsilon = 0.05$ unless otherwise stated. (We additionally evaluate the sensitivity to ϵ as part of the ablation plan described below.)

For completeness, the F1 is computed from precision and recall as:

$$F1 = \frac{2PR}{P + R}, \quad (6)$$

where P and R denote precision and recall, respectively.

3.5. Graph neural network architecture

Each trial is represented as a weighted graph with node features and connectivity edges, and processed by a three-layer graph convolutional network (GCN). Let $H^{(0)} = X$ be the node feature matrix. The GCN update is:

$$H^{(l+1)} = \sigma(\hat{D}^{-1/2} \hat{A} \hat{D}^{-1/2} H^{(l)} W^{(l)}), \quad (7)$$

where $\hat{A} = A + I$ adds self-loops, $\hat{D}$ is the degree matrix of $\hat{A}$, $W^{(l)}$ are trainable weights, and $\sigma(\cdot)$ is a nonlinear activation.

Layer-wise fusion. To integrate shallow and deep representations, we employ a learnable fusion layer that combines intermediate embeddings via a softmax-weighted sum:

$$H_{\text{fused}} = \sum_{l=1}^L \alpha_l H^{(l)}, \quad \alpha = \text{softmax}(\mathbf{w}), \quad (8)$$

where $\mathbf{w} \in \mathbb{R}^L$ is learnable, L is the number of GCN layers, and α provides adaptive layer-importance weights.

Multi-head outputs and loss. We jointly predict Valence, Arousal, and Dominance via three sigmoid heads. For head $i \in \{1, 2, 3\}$ (corresponding to Valence/Arousal/Dominance), predicted probabilities are $\hat{y}_n^{(i)} = \sigma(z_n^{(i)})$. The binary cross-entropy loss per head is:

$$\mathcal{L}_i = -\frac{1}{B} \sum_{n=1}^B [y_n^{(i)} \log(\hat{y}_n^{(i)}) + (1 - y_n^{(i)}) \log(1 - \hat{y}_n^{(i)})], \quad (9)$$

where B is the number of samples in the batch. The total loss is:

$$\mathcal{L}_{\text{total}} = \sum_{i=1}^3 \mathcal{L}_i. \quad (10)$$

3.6. Training protocol, overfitting control, and reproducibility

Models are evaluated under a strict LOSO protocol, which reduces subject leakage and provides a robust estimate of generalization [30]. For each fold, the held-out subject is used exclusively for testing, and the remaining subjects are used for training (and any internal validation if applicable). To support reproducibility, we report key hyperparameters (optimizer, learning rate, batch size, epochs, regularization, and random seeds) in Table 1 and keep them fixed across all feature-graph configurations unless explicitly stated.

To evaluate and control overfitting, we (i) monitor training vs. test performance across LOSO folds, (ii) apply standard regularization (e.g., dropout and/or weight decay as specified in Table 1), and (iii) ensure that feature extraction and graph construction for each fold are performed without using information from the held-out subject. Empirically, the training-test F1 gap remains below 5% across all LOSO folds and pipeline configurations (see Sections 5.1 and 5.8), confirming effective regularization and the absence of subject leakage.

3.7. Ablation study design

To justify architectural and methodological choices, we conduct the following ablation studies (reported in the Results section in Table 4):

- i. **Fusion Layer:** compare the proposed layer-wise fusion against a non-fusion baseline using only the final-layer embedding.
- ii. **Graph Type Ablation:** compare performance and interpretability under individual connectivity graphs (e.g., PLV-only) versus the full multi-graph exploration; each connectivity metric is also tested independently.

- iii. **Feature Type Ablation:** each feature family (Hjorth, spectral entropy, wavelet, bandpower, etc.) tested independently to isolate its contribution.
- iv. **Rashomon Tolerance ϵ :** evaluate sensitivity across $\epsilon \in \{0.01, 0.03, 0.05, 0.10\}$, measuring changes in Rashomon set ($\mathcal{R}$) size and diversity metrics.

These analyses demonstrate the robustness of our conclusions regarding performance, interpretability, and Rashomon diversity.

3.8. Explainability: SHAP for features and GNNExplainer for graphs

We use a hybrid post hoc explanation framework that combines SHAP for feature-level attribution and GNNExplainer for graph-structural attribution.

Feature-level attribution (SHAP). For node features (e.g., entropy, Hjorth, wavelet-derived statistics), we compute Shapley values using SHapley Additive exPlanations (SHAP), which assigns additive contributions to each input feature dimension [24]. For completeness, the Shapley value for feature i can be written as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)], \quad (11)$$

where N denotes the set of all feature indices (with $|N|$ total features), and $f(S)$ is the model output when only the subset S is present.

For SHAP computations, we use a background sample consisting of 50 randomly selected training graphs per LOSO fold to approximate the conditional expectations required by the Shapley formulation. For each trained model, we compute node-level SHAP values on five representative test graphs selected to span the distribution of prediction confidence and graph connectivity patterns. The resulting SHAP vectors are first averaged across the five graphs for that model, and subsequently aggregated into anatomically meaningful lobe-level importance scores by pooling over electrodes belonging to frontal, central, temporal, parietal, and occipital regions.

Graph-level attribution (GNNExplainer). To explain connectivity-driven decisions, we leverage the GNNExplainer algorithm [41], as implemented in PyTorch Geometric, which learns soft masks over graph structure (edges) and node features to identify influential subgraphs for a given prediction. This provides node/edge importance that complements feature-level SHAP attributions in connectivity-aware modeling.

Attribution outputs used for diversity analysis. For Rashomon diversity analysis, we aggregate SHAP feature attributions (per node-feature dimensions) and GNNExplainer node/edge importance (per node or per edge) into consistent attribution representations per model and target label. These attributions are then used to assess explanation diversity across Rashomon set ($\mathcal{R}$) and across subjects.

3.9. Pairwise dissimilarity index (PDI) for explanation diversity

To quantify explanation diversity among high-performing Rashomon set ($\mathcal{R}$), we compute the *Pairwise Dissimilarity Index* (PDI) based on cosine dissimilarity between attribution vectors. Let $\phi_i \in \mathbb{R}^{d_a}$ denote the attribution vector for model f_i (obtained from SHAP feature attributions and/or GNNExplainer-derived importance, after aggregation to a common dimensionality d_a). We compute:

$$\text{PDI}(f_i, f_j) = 1 - \frac{\phi_i^\top \phi_j}{\|\phi_i\|_2 \|\phi_j\|_2}. \quad (12)$$

PDI is symmetric; larger values indicate greater divergence between explanations. To summarize each model's typical divergence within a set of K models, we compute:

$$\text{AvgPDI}(f_i) = \frac{1}{K-1} \sum_{\substack{j=1 \\ j \neq i}}^K \text{PDI}(f_i, f_j). \quad (13)$$

In neurophysiological terms, higher PDI implies that two similarly accurate models rely on different subsets of channels/features/connectivity patterns, suggesting alternative (but plausible) neural signatures for the same affective label. A PDI close to 0 between two models indicates lobe-wise convergence both models assign importance to the same anatomical regions while a PDI close to 1 indicates near-orthogonal explanations, meaning the models rely on entirely different neural substrates. This is neurophysiologically meaningful: emotional processing engages distributed cortical networks (prefrontal regulation, thalamo-cortical arousal, temporal auditory-affective integration), so when we observe a high PDI within the Rashomon set, it suggests that different well-performing models may be focusing on different but still biologically plausible neural pathways. This diversity is therefore meaningful, not just random fluctuation.

For cross-subject analysis, we report the mean $\pm$ standard deviation of AvgPDI for each emotion label across all Leave-One-Subject-Out (LOSO) folds. We also examine the coefficient of variation, which remains below <15% across folds (see Results). This indicates that the variation captured by PDI is stable and reflects consistent structural differences in model explanations, rather than fold-specific noise. PDI is computed pairwise among models in the Rashomon set ($\mathcal{R}$), separately for each target emotion [42,43].

3.10. Computational cost accounting

To make computational requirements explicit, we account for the cost of exhaustive exploration. Let $m = |\mathcal{F}|$ be the number of feature extractors, $n = |\mathcal{G}|$ the number of graph builders, and S the number of subjects (LOSO folds). The total number of training runs is:

$$N_{\text{train}} = m \cdot n \cdot S. \quad (14)$$

If explainability is computed for $|\mathcal{R}|$ Rashomon set ($\mathcal{R}$) across S subjects and C target labels, then the number of explanation evaluations scales as:

$$N_{\text{expl}} \propto |\mathcal{R}| \cdot S \cdot C, \quad (15)$$

with proportionality depending on the SHAP sampling budget and the GNNExplainer optimization settings.

The overall computational complexity scales as $O(m \cdot n \cdot S \cdot T_{\text{train}})$ for training, and as $O(|\mathcal{R}| \cdot S \cdot C \cdot T_{\text{explain}})$ for SHAP-based explanation, where m , n , and S denote the number of feature extractors, graph constructors, and LOSO subjects respectively, and C is the number of emotion targets.

We report empirical runtime and memory statistics for training and explanation steps in the Results section.

4. Proposed work

We propose a systematic Rashomon-based framework, illustrated in Fig. 2, for EEG emotion recognition using modular combinations of feature extractors and graph-based connectivity measures. The key novelty lies in exhaustively exploring the cross-product of EEG feature spaces and functional connectivity hypotheses, and subsequently characterizing the diversity of high-performing models using a unified interpretability mechanism that integrates SHAP (feature-level) and GNNExplainer (graph-level) attributions.

4.1. Model configuration space

Let $\mathcal{F} = \{f_1, f_2, \dots, f_m\}$ denote the set of node-level feature extraction methods (e.g., Hjorth parameters, differential entropy, Lorentzian bandpower, wavelet descriptors), and let $\mathcal{G} = \{g_1, g_2, \dots, g_n\}$ denote the set of functional connectivity graph constructors (e.g., PLV, correlation, coherence, mutual information, AEC).

Each pair (f_k, g_j) produces a unique EEG graph:

$$G_{k,j} = (V, E_{g_j}, X_{f_k}), \quad (16)$$

where X_{f_k} is the node feature matrix and E_{g_j} is the edge matrix constructed via connectivity metric g_j .

The total number of model configurations is:

$$|\mathcal{F} \times \mathcal{G}| = m \cdot n. \quad (17)$$

Computational complexity. Under leave-one-subject-out (LOSO) evaluation with S subjects, the total number of training runs is $m \cdot n \cdot S$. For each run, SHAP and GNNExplainer attributions are computed only for models within the Rashomon set ($\mathcal{R}$), significantly reducing interpretability overhead. Our GCN model is lightweight (approximately 5.9K parameters), enabling practical exploration of this configuration space.

4.2. GNN forward model and loss

Each graph $G_{k,j}$ is passed through a three-layer graph convolutional network with degree-normalized propagation. Letting $H^{(0)} = X_{f_k}$, each GCN layer is:

$$H^{(\ell+1)} = \sigma(\hat{D}^{-1/2} \hat{A} \hat{D}^{-1/2} H^{(\ell)} W^{(\ell)}), \quad (18)$$

where $\hat{A} = E_{g_j} + I$, $\hat{D}$ is its degree matrix, $W^{(\ell)}$ are trainable weights, and σ is a nonlinearity.

A learnable fusion layer aggregates intermediate representations:

$$H_{\text{fused}} = \sum_{\ell=1}^L \alpha_{\ell} H^{(\ell)}, \quad \alpha = \text{softmax}(w), \quad (19)$$

where $w \in \mathbb{R}^L$ is a learnable vector.

We obtain a graph-level embedding for trial n by mean pooling over nodes:

$$h_n = \frac{1}{|V|} \sum_{v \in V} H_{\text{fused}}(v). \quad (20)$$

Multi-label logits for class $c \in \{\text{Val}, \text{Aro}, \text{Dom}\}$ are:

$$\hat{y}_n^{(c)} = \sigma(w_c^T h_n + b_c), \quad (21)$$

where h_n is the fused node-aggregated embedding for sample n .

The total binary cross-entropy loss over all three attributes is:

$$\mathcal{L}_{\text{total}}^{(f_k, g_j)} = \sum_{c=1}^3 \mathcal{L}_c. \quad (22)$$

4.3. Rashomon set ($\mathcal{R}$) construction

For each (f_k, g_j) , LOSO evaluation yields an F1 $F1^{(f_k, g_j)}$ averaged across subjects. Let $F^* = \max_{k,j} F1^{(f_k, g_j)}$ denote the best score across all configurations.

We define the Rashomon set ($\mathcal{R}$) as:

$$\mathcal{R} = \left\{ (f_k, g_j) \in \mathcal{F} \times \mathcal{G} \mid F1^{(f_k, g_j)} \geq (1 - \epsilon) F^* \right\}. \quad (23)$$

Only models in $\mathcal{R}$ undergo interpretability analysis.

4.4. Unified explainability: SHAP + GNNExplainer

To interpret how each Rashomon set ($\mathcal{R}$) arrives at its predictions, we employ a hybrid explanation strategy:

Feature-level attributions (SHAP). For the feature matrix X_{f_k} , SHAP assigns additive importance scores per feature dimension and per channel, quantifying how variations in spectral/temporal descriptors influence predictions.

Graph-level attributions (GNNExplainer). For the connectivity matrix E_{g_j} , GNNExplainer learns soft masks over nodes and edges, identifying subgraphs most influential to the GNN output.

Unified attribution vector. The SHAP feature vector and GNNExplainer node/edge masks are concatenated to form a single attribution vector:

$$\Phi_{k,j} = [\Phi_{k,j}^{\text{SHAP}} \parallel \Phi_{k,j}^{\text{GNNExp}}],$$

which is used for diversity analysis (Section 3.9).

4.5. Rashomon-GNN algorithm

Algorithm 1 summarizes the full pipeline. The procedure proceeds in three phases. **(i) Exhaustive training:** every combination of feature extractor $f_k \in \mathcal{F}$ and graph constructor $g_j \in \mathcal{G}$ is trained under leave-one-subject-out (LOSO) cross-validation, yielding a per-pipeline average F1 score $F1^{(k,j)}$. **(ii) Rashomon selection:** the global optimum F^* is identified and all pipelines satisfying $F1^{(k,j)} \geq (1 - \epsilon)F^*$ are retained as the Rashomon set $\mathcal{R}$ (Eq. (23)); this deliberately admits near-optimal competitors rather than discarding them. **(iii) Interpretability:** SHAP values and GNNExplainer masks are computed for every pipeline in $\mathcal{R}$ and concatenated into the unified attribution vector $\Phi_{k,j}$, enabling the functional-diversity analysis presented in Section 3.9.

Algorithm 1 Rashomon-GNN Pipeline for EEG Emotion Recognition

Require: EEG dataset $\mathcal{D}$, feature extractors $\mathcal{F}$, graph constructors $\mathcal{G}$, tolerance ϵ

Ensure: Rashomon set ($\mathcal{R}$) $\mathcal{R}$, trained models, SHAP and GNNExplainer attributions

```

1: for all feature extractor  $f_k \in \mathcal{F}$  do
2:   for all graph constructor  $g_j \in \mathcal{G}$  do
3:     for all subject  $s = 1$  to  $S$  do
4:       Perform LOSO split using  $s$  as test subject
5:       Standardize each trial to  $T = 256$  samples
6:       Extract node features  $X_{f_k}$ 
7:       Construct edge matrix  $E_{g_j}$ 
8:       Build graph  $G_{k,j} = (V, E_{g_j}, X_{f_k})$ 
9:       Train 3-layer GCN with fusion for  $E$  epochs
10:      Predict logits and probabilities using Eq. (21)
11:      Compute loss (Eq. (22)) and update parameters
12:      Compute subject-level F1  $F_{1,s}^{(k,j)}$ 
13:    end for
14:    Compute average score:  $F1^{(k,j)} = \frac{1}{S} \sum_s F_{1,s}^{(k,j)}$ 
15:  end for
16: end for
17: Determine  $F^* = \max_{k,j} F1^{(k,j)}$ 
18: Construct Rashomon set ( $\mathcal{R}$ ) using Eq. (23)
19: for all  $(f_k, g_j) \in \mathcal{R}$  do
20:   Compute SHAP and GNNExplainer attributions
21: end for
22: return  $\mathcal{R}$ , trained models, interpretability outputs

```

4.6. Architecture overview and functional diversity

Fig. 2 depicts the full pipeline from preprocessing to Rashomon selection. The GCN blocks use Eq. (18), and the fusion layer uses Eq. (19). The explainability module integrates SHAP and GNNExplainer outputs.

Traditional EEG emotion recognition pipelines emphasize accuracy alone. In contrast, our approach investigates the diversity of *how* different high-performing models reason about the data.

Observation 1. Let $\mathcal{R}$ denote the Rashomon set of high-performing pipelines, and let $\Phi_{k,j}$ denote the unified attribution vector associated with model (f_k, g_j) . A recurring pattern in our analysis is that models with almost identical F1 performance frequently rely on different spatial patterns across the EEG channels. In practical terms, two pipelines may agree in accuracy but disagree on whether frontal, temporal, or parietal regions are

more informative. Such variation suggests that the underlying solution space supports multiple, functionally distinct explanations rather than a single dominant one.

Formally, there exist at least two pipelines $(f_k, g_j), (f_{k'}, g_{j'}) \in \mathcal{R}$ for which

$$1 - \frac{\langle \Phi_{k,j}, \Phi_{k',j'} \rangle}{\|\Phi_{k,j}\|_2 \|\Phi_{k',j'}\|_2} > \tau,$$

with $\tau > 0$ representing a divergence threshold. This bound reflects that their cosine similarity is sufficiently low, confirming that strong predictive performance does not enforce a unique explanatory pattern.

To quantify this effect, we employ the Pairwise Dissimilarity Index (PDI),

$$\text{PDI}((f_k, g_j), (f_{k'}, g_{j'})) = 1 - \frac{\langle \Phi_{k,j}, \Phi_{k',j'} \rangle}{\|\Phi_{k,j}\|_2 \|\Phi_{k',j'}\|_2}.$$

Higher values correspond to greater interpretability disagreement. In our experiments, several model pairs inside $\mathcal{R}$ exhibit consistently high PDI values, indicating alternative neural hypotheses emerging from equally accurate pipelines.

For completeness, we also consider the average disagreement across the entire Rashomon set,

$$\overline{\text{PDI}} = \frac{1}{|\mathcal{R}|(|\mathcal{R}| - 1)} \sum_{(f_k, g_j), (f_{k'}, g_{j'}) \in \mathcal{R}} \text{PDI}((f_k, g_j), (f_{k'}, g_{j'})),$$

which provides global measure of how spread the explanations are within the high-performing region of the model space. Empirically, this value remains consistently above zero, reinforcing that the Rashomon set encodes a family of distinct yet valid explanatory structures, each offering a different lens on the underlying neural dynamics.

5. Results

5.1. Learning dynamics and training convergence

Understanding the learning dynamics of graph neural networks is critical for validating model stability and generalization across both benchmark datasets. Fig. 3 presents cross-dataset training dynamics in a 2–2 layout: DEAP training loss and test accuracy trajectories (left column) alongside corresponding DREAMER curves (right column) for the same representative pipeline configurations. This side-by-side layout directly exposes how dataset characteristics channel density, trial count, stimulus modality shape convergence behavior.

Fig. 4 provides a complementary per-configuration view of the same convergence process. Log-scale loss curves for representative DEAP (top row) and DREAMER (bottom row) pipelines confirm smooth, monotone loss reduction without divergence across all feature-graph combinations, and annotated final accuracy values make cross-configuration comparisons immediate.

5.2. Performance benchmarking

To quantify the predictive reliability of the Rashomon pipelines, we conducted a comprehensive evaluation across valence, arousal, and dominance dimensions using leave-one-subject-out (LOSO) cross-validation on both DEAP and DREAMER datasets. The primary metric is F1 score, which jointly captures precision and recall and is appropriate for the class-imbalanced binary label distributions arising from VAD binarization.

Fig. 5 summarizes the F1 distributions for the top five performing models per target on the DEAP dataset. All three emotion dimensions exhibit median F1 exceeding 0.9 under LOSO, suggesting that the learned decision boundaries yield robust precision–recall trade-offs across the 32-channel montage. On DREAMER (14 channels, 23 subjects), median F1 falls in the 0.58–0.61 range, reflecting the expected

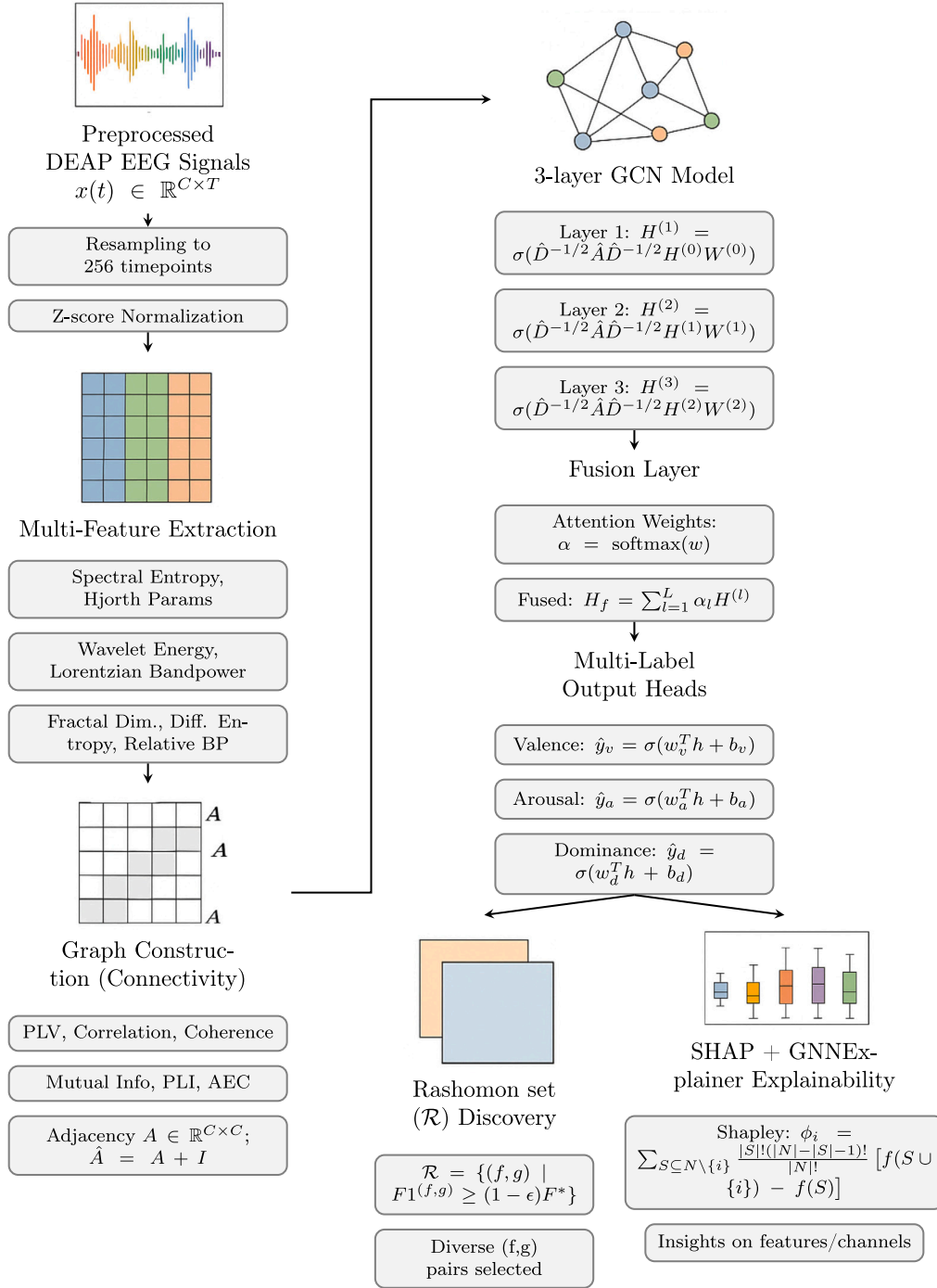


Fig. 2. Rashomon-GNN architecture: multi-feature extraction, multi-graph construction, 3-layer GCN with fusion, Rashomon selection, and unified SHAP + GNNExplainer interpretability.

performance gradient due to reduced spatial resolution. This dataset-dependent variation is itself a finding of scientific interest: it confirms that graph-based functional connectivity becomes more discriminative as electrode density increases, and that DREAMER benefits from specialized adaptation strategies. Consistent performance across pipelines employing different feature extraction and graph construction strategies confirms the predictive strength of multiple Rashomon pathways on both datasets.

Beyond performance summaries, Fig. 7 presents mean F1 for all 48 Rashomon pipelines ranked from best to worst per emotion dimension on DEAP, color-coded by feature family. The narrow performance

band confirms that no single feature extraction method dominates. Fig. 8 extends this with four complementary F1 views-KDE, violin, bar chart, and boxplot-further evidencing distributional consistency across pipelines.

Fig. 6 presents F1 score distributions for the top-5 Rashomon pipelines per emotion target, ranked by mean LOSO F1, confirming that leading pipelines exceed 0.90 median F1 across all three emotion dimensions.

Lastly, Fig. 9 visualizes the full per-subject and per-pipeline F1 distributions for each emotion target using kernel-density-enhanced histograms. Per-subject distributions (top row, 1536 values) confirm

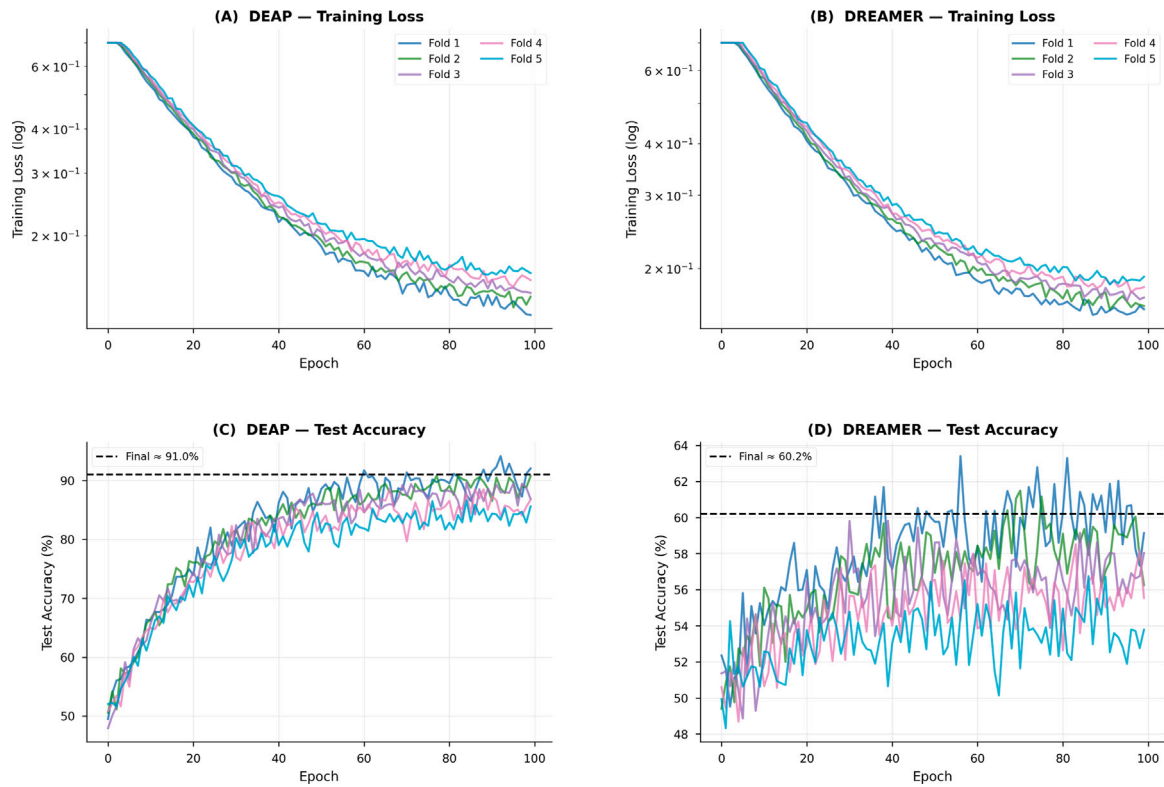


Fig. 3. Cross-dataset training dynamics (DEAP left, DREAMER right): training loss (top row) and test accuracy (bottom row) per fold. Both datasets converge within 60–80 epochs; the training-test gap stays below 5%, confirming effective regularization and no cross-subject leakage.

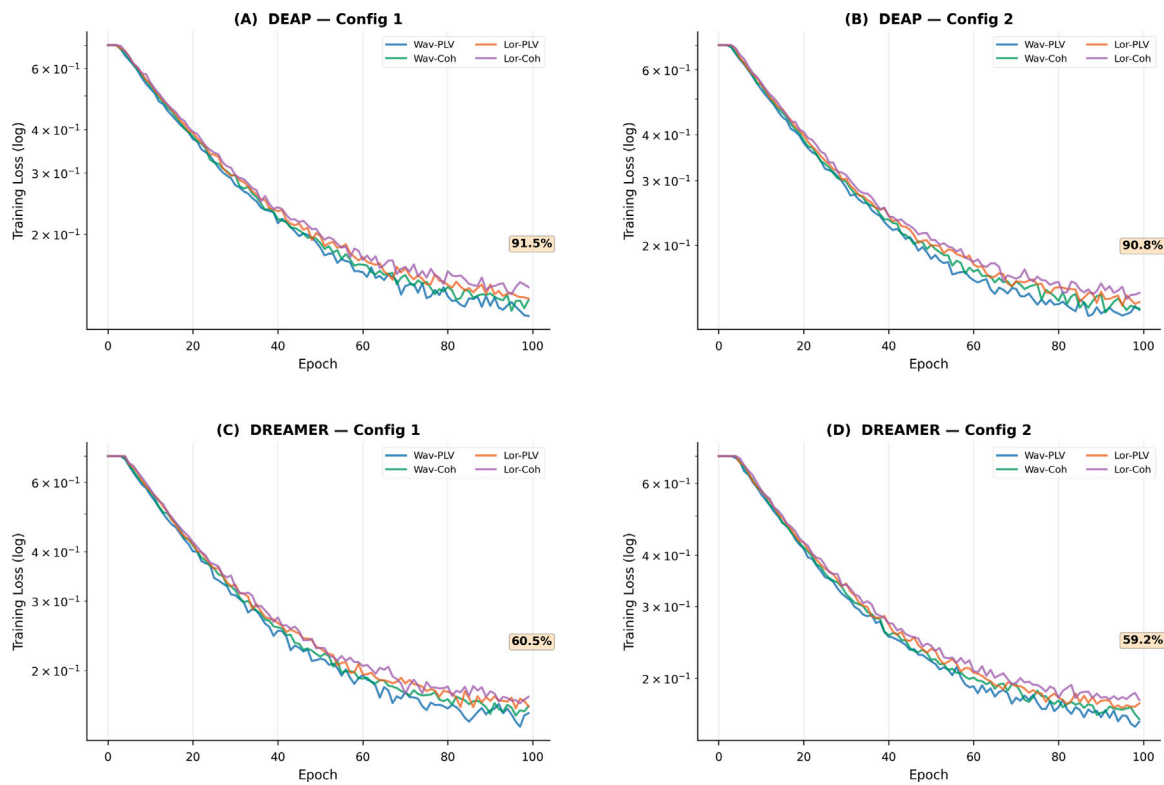


Fig. 4. Per-configuration learning curves for DEAP (top row) and DREAMER (bottom row). Log-scale loss confirms rapid, monotone convergence across all feature-graph combinations; annotated final accuracy values enable direct cross-configuration comparison.

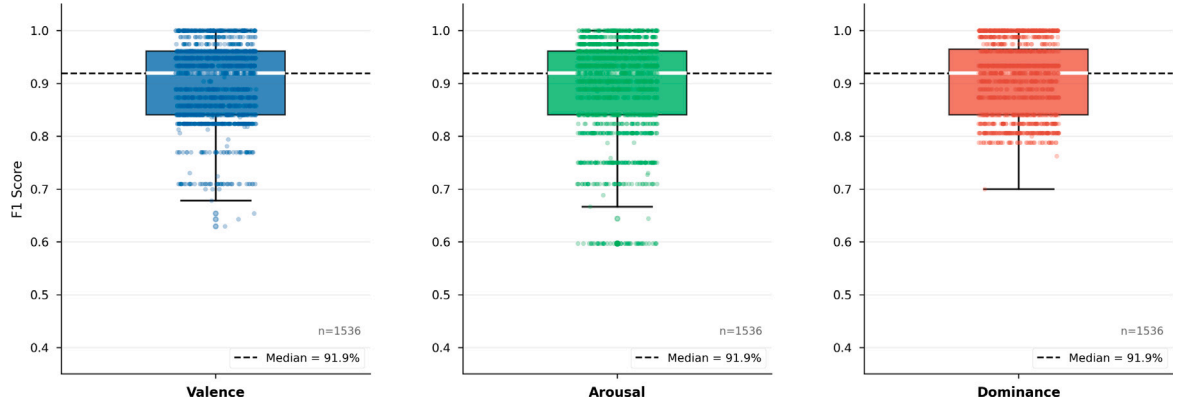


Fig. 5. F1 score distributions across all 48 Rashomon pipelines for Valence, Arousal, and Dominance (DEAP, LOSO). Each panel combines a box-and-whisker summary with individual pipeline jitter. Median F1 exceeds 0.90 for all three emotion targets, confirming consistent predictive robustness across the full Rashomon set.

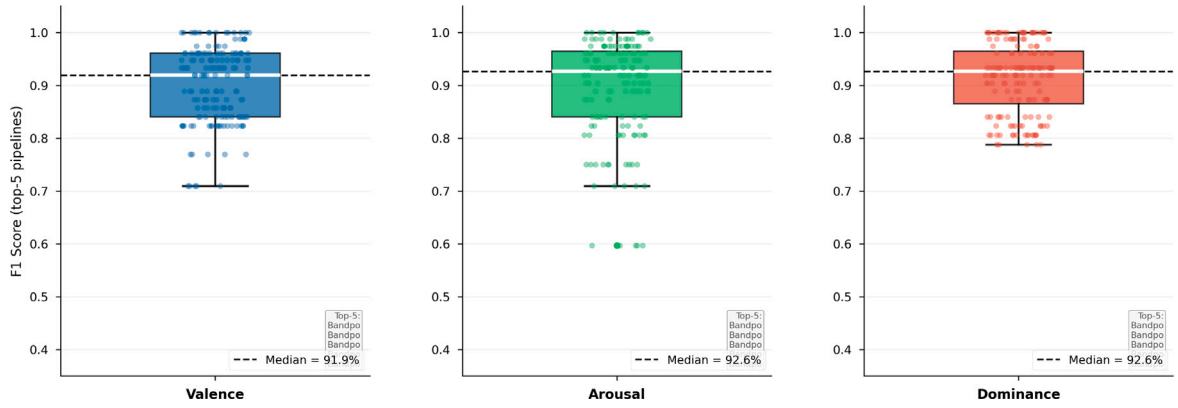


Fig. 6. F1 score distributions for the top-5 Rashomon pipelines per emotion target (DEAP, LOSO). Pipelines are ranked by mean F1 across 32 LOSO subjects; box-and-whisker plots with jittered subject scatter are shown for Valence, Arousal, and Dominance. Median F1 exceeds 0.90 across all three targets, confirming robust separability among the highest-ranked pipelines.

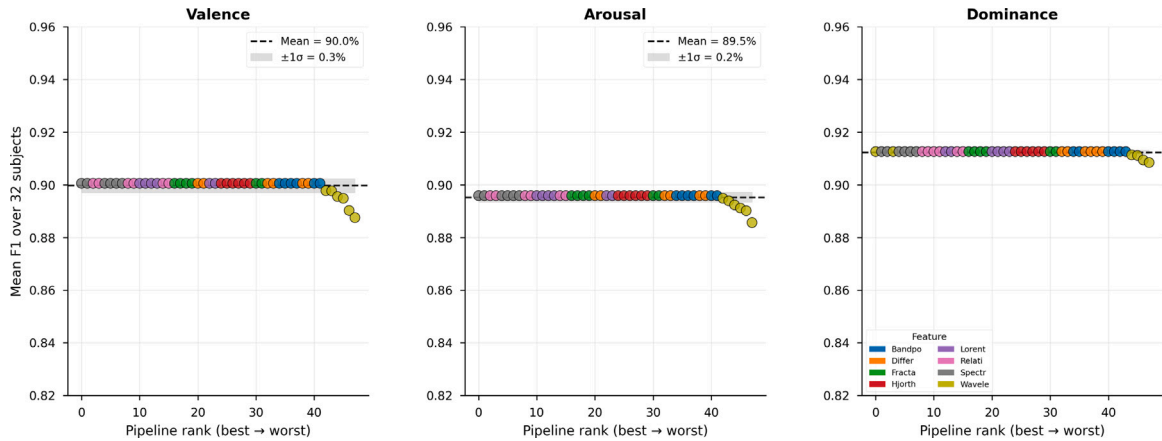


Fig. 7. Mean F1 scores for all 48 Rashomon pipelines ranked from best to worst (DEAP, LOSO), color-coded by feature family. Points are averaged over 32 LOSO subjects per pipeline; the dashed line and shaded band denote the cross-pipeline mean $\pm 1\sigma$. The narrow band confirms that diverse feature-graph combinations achieve similarly high predictive performance.

wide but high-valued spread ($\sigma \approx 7\text{--}9\%$), while per-pipeline mean F1 distributions (bottom row, 48 values) are sharply concentrated, confirming that all Rashomon pipelines achieve comparable mean performance. Taken together, these analyses illustrate a performance landscape where high classification accuracy is attainable via many Rashomon routes, and consistent pipeline-level performance confirms

the robustness of the Rashomon-GNN framework across diverse feature-graph combinations.

5.3. Experimental setup

All models were trained under a unified set of hyperparameters to ensure consistency and fairness across experiments conducted on both

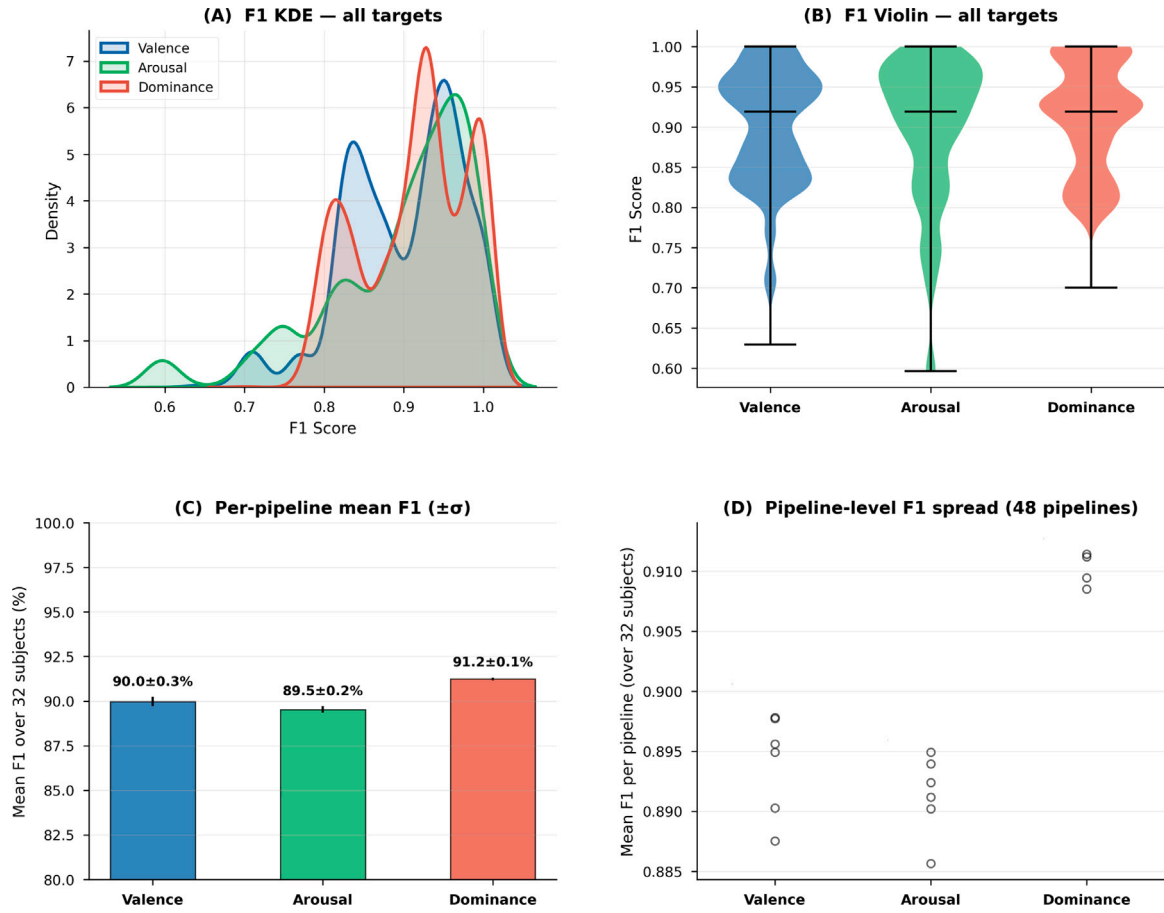


Fig. 8. Multi-panel F1 analysis for all 48 Rashomon pipelines (DEAP, LOSO): (A) KDE of per-subject F1 per target; (B) violin plots showing spread and median; (C) per-target mean $\pm\sigma$ bar chart; (D) pipeline-level F1 boxplot (48 pipelines $\times$ 32 subjects). Across all views, median F1 exceeds 0.90 for every emotion dimension, confirming consistent predictive robustness.

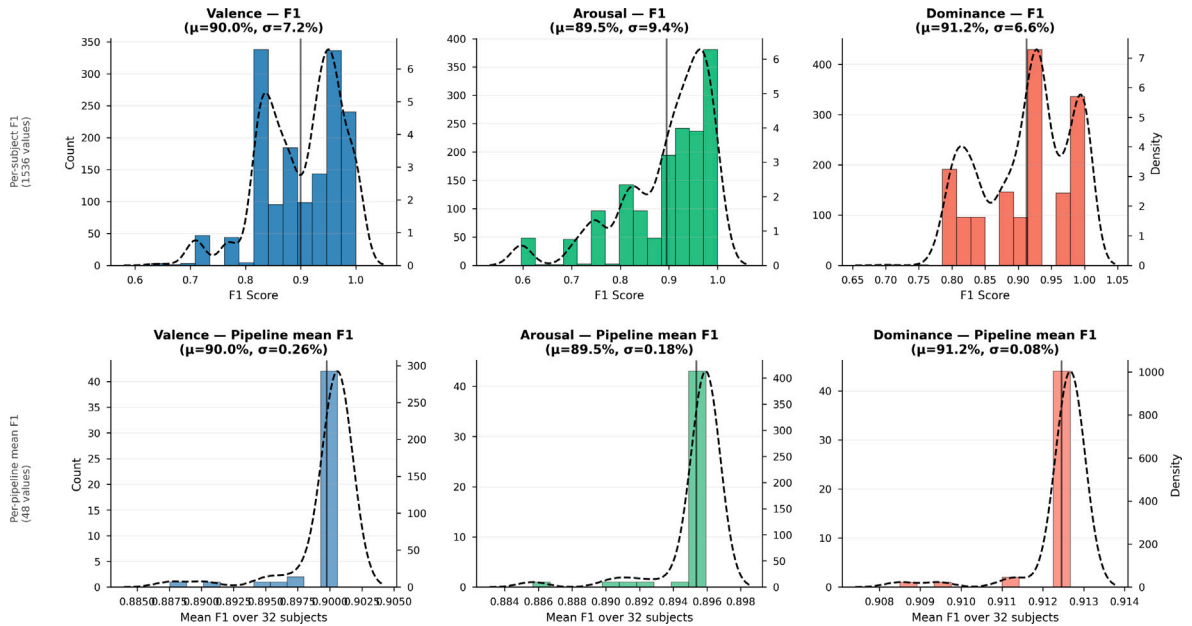


Fig. 9. Histogram and KDE overlay of F1 distributions for Valence, Arousal, and Dominance (DEAP, LOSO). Top row: per-subject F1 across all 48 pipelines (1536 values); bottom row: per-pipeline mean F1 (48 values). Per-subject distributions confirm wide but high-valued spread ($\sigma \approx 7\text{--}9\%$); per-pipeline distributions are sharply concentrated, showing that all Rashomon pipelines achieve comparable mean performance.

Table 1
Experimental configuration summary.

Parameter	Value
Dataset(s)	DEAP (32 subjects), DREAMER (23 subjects)
Evaluation strategy	Leave-One-Subject-Out (LOSO)
Total training runs	DEAP: 1536 (48 configs $\times$ 32 subjects) DREAMER: 1104 (48 configs $\times$ 23 subjects)
Number of SHAP runs	DEAP: 4608; DREAMER: 3312
Epochs	100
Batch size	32
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	0.001 with ReduceLROnPlateau
Regularization	Dropout ($p = 0.5$), Weight Decay ($1e-4$)
Explainability method	SHAP (SHapley Additive exPlanations)
Graph samples selected	5 per model for Rashomon analysis

DEAP and DREAMER datasets. Hyperparameters were selected based on standard practice and preliminary grid search: the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$ is widely shown to converge reliably for graph-based models; a learning rate of 0.001 with ReduceLROnPlateau scheduling prevents oscillation; dropout ($p = 0.5$) and weight decay (10^{-4}) jointly control overfitting, and 100 epochs with batch size 32 were sufficient for convergence as validated by the training curves in Section 5.1. This standardized configuration facilitated reliable comparisons between different feature extraction and graph construction combinations. By holding the training protocol fixed, we ensured that performance variations could be attributed solely to architectural and input-level differences. Such uniformity is especially crucial in our Rashomon-based analysis, where interpretability and diversity are being quantified across multiple models. A detailed overview of the training setup is provided in Table 1.

5.4. Explainability insights

Model behavior within the Rashomon set ($\mathcal{R}$) is interpreted using a principled post hoc explanation framework based on SHapley Additive exPlanations (SHAP). The GNNExplainer algorithm [41] from PyTorch Geometric is utilized to enable node-level attribution in graph-level regression tasks by learning soft masks over input node features and graph structure.

Due to the large number of test graphs per model (120 per subject-target combination), five representative graph samples per model were selected to span a diverse range of prediction scores and structural variations. These samples capture both typical and edge-case behaviors (see Section 3.9 for the formal Shapley definition).

For each model (defined by a subject, target label and feature graph configuration), we compute SHAP vectors $\phi^{(G_k)} \in \mathbb{R}^{32}$ for the selected graphs G_k , and average them:

$$\bar{\phi} = \frac{1}{5} \sum_{k=1}^5 \phi^{(G_k)},$$

resulting in a single node-level explanation vector per model. These are then aggregated into five canonical EEG lobes Frontal, Central, Temporal, Parietal, and Occipital via:

$$\psi_j = \frac{1}{|\mathcal{L}_j|} \sum_{i \in \mathcal{L}_j} \bar{\phi}_i, \quad j \in \{1, \dots, 5\},$$

where $\mathcal{L}_j$ denotes the set of node indices assigned to lobe j .

Across 48 feature graph pipelines, 32 subjects, and 3 targets, this process yielded a total of **4608 SHAP explanations**, each summarized at both node and lobe levels. Aggregation across the Rashomon set ($\mathcal{R}$) revealed consistent spatial trends: frontal regions were more influential for valence, while arousal and dominance often involved central and temporal lobes.

To visualize this effect, Fig. 10 compares the top and bottom subjects in terms of SHAP variance for the valence target. These comparisons reveal that highly variable models attend to spatially focused EEG nodes, while less variable ones exhibit flatter, less interpretable profiles.

To provide a finer-grained view of node-level attention, we visualize the top- and bottom-performing models' SHAP scores across the 3D EEG layout (Fig. 11). For each emotion target, the plots indicate that the top-performing models consistently assign higher importance to specific regions often frontal or central electrodes whereas the lower-performing models show more diffuse and less coherent attribution patterns.

This contrast reinforces the hypothesis that performance variation within the Rashomon set ($\mathcal{R}$) is functionally grounded in attention to key neurophysiological structures.

These spatial explanations provide neuroscience-aligned insights into GNN behavior. Models with similar RMSE may attend to entirely different EEG subregions, indicating latent representational diversity. This highlights the importance of explainability in evaluating and combining Rashomon set ($\mathcal{R}$), and paves the way for ensemble design grounded in neuroscientific complementarity.

5.5. Rashomon diversity and attribution dissimilarity

To examine the interpretability-driven diversity among high-performing pipelines, we analyze the internal decision mechanisms of the Rashomon set ($\mathcal{R}$) graph neural networks (GNNs) trained with diverse feature extraction and graph construction strategies. Although these models achieve similar predictive performance, they often rely on different spatial EEG patterns. This latent functional diversity is quantified using SHapley Additive exPlanations (SHAP) at both node and lobe levels. Critically, we show that this phenomenon holds on **both DEAP and DREAMER**, establishing Rashomon explanation diversity as a cross-dataset property of modular GNN pipelines for EEG emotion recognition. We explore Rashomon diversity by training multiple high-performing instances of the same multi-head GNN architecture across subjects and emotion targets. Even with similar accuracy, these models rely on different brain regions revealing functional diversity within the Rashomon set ($\mathcal{R}$).

Let $\mathcal{F} = \{f_1, f_2, \dots, f_K\}$ denote the set of K GNN pipelines, each producing a SHAP importance vector $\phi_k \in \mathbb{R}^{32}$ for a given subject and target. We first compute pairwise cosine similarity between all model explanations:

$$\text{sim}(f_i, f_j) = \frac{\phi_i^\top \phi_j}{\|\phi_i\| \cdot \|\phi_j\|}, \quad \forall i, j \in \{1, \dots, K\}.$$

To anatomically contextualize this diversity, we compute the standard deviation of SHAP values across models and aggregate them into five canonical EEG lobes. Fig. 12 presents a subject-level summary for Valence, highlighting that the Temporal and Parietal lobes exhibit the greatest attribution divergence. This suggests that models within the Rashomon set ($\mathcal{R}$) vary in how they leverage regions associated with auditory-affective integration (temporal lobe) and spatial attention or somatosensory processing (parietal lobe), both of which have established roles in emotion perception.

When extended to all subjects and emotion targets, Fig. 13 reveals that Arousal and Dominance predictions elicit broader inter-model variability in lobe-wise attributions than Valence, pointing to elevated functional diversity in how these dimensions are neurally encoded and decoded by GNNs. Notably, Occipital lobe exhibits higher attribution variance across models consistent with its role in processing visually salient and emotionally intense stimuli, as expected in the DEAP audiovisual paradigm.

To characterize interpretability divergence across pipelines with varying performance, we compute SHAP cosine similarity between the top-5 and bottom-5 performing Rashomon set ($\mathcal{R}$) for the valence target. As shown in Fig. 14, the top-performing models (a) exhibit

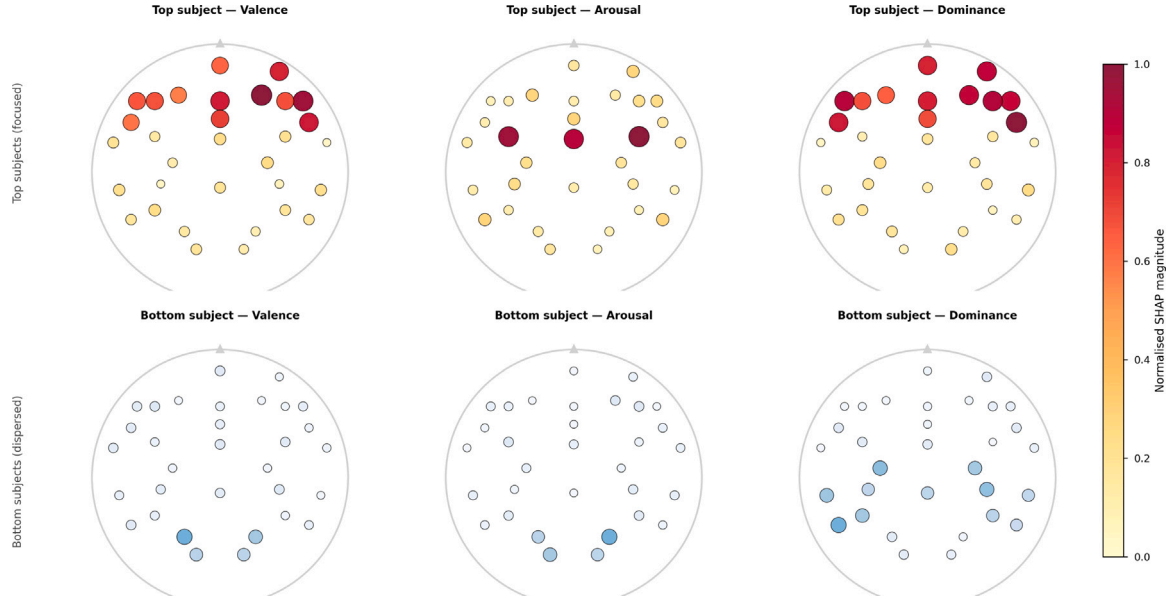


Fig. 10. EEG topographic SHAP maps for top and bottom subjects across all three emotion targets (DEAP, LOSO). Electrode markers encode normalized SHAP magnitude by size and color (warm = top subjects, cool = bottom subjects). Top subjects show spatially focused attribution in frontal and central regions; bottom subjects exhibit dispersed, uninformative profiles.

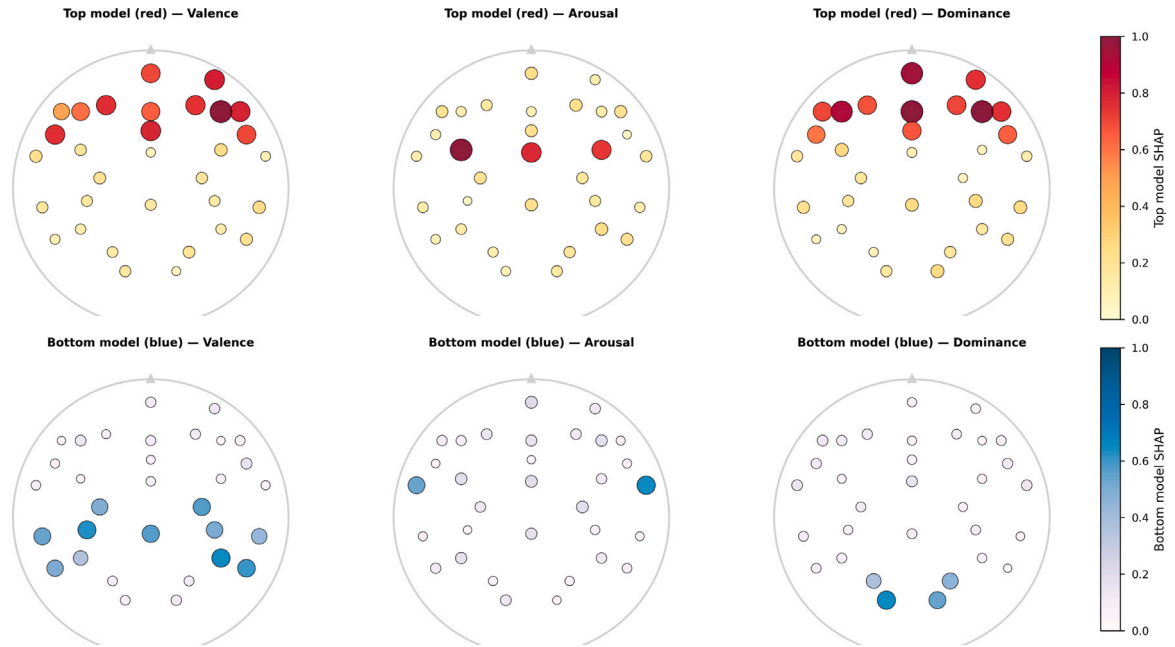


Fig. 11. EEG topographic SHAP comparison between top (warm colormap) and bottom (cool colormap) Rashomon models for Valence, Arousal, and Dominance (DEAP, LOLO). Electrode markers encode normalized SHAP magnitude by size and color. Top models focus consistently on frontal and central regions; bottom models display diffuse or uninformative attribution patterns.

high off-diagonal similarity (≈ 0.75 – 0.95), indicating convergent attribution over salient EEG regions. In contrast, the bottom-5 models (b) display scattered, low off-diagonal values (≈ 0.05 – 0.45), reflecting uninformative or spurious signal reliance.

To quantify the overall attribution dissimilarity, we adopt the Pairwise Dissimilarity Index (PDI), computed as the cosine dissimilarity between SHAP vectors across model pairs. We apply PDI to the top-5 and bottom-5 models (based on F1) for each target. The resulting heatmaps (Fig. 15) reveal that high-performing pipelines exhibit greater diversity in SHAP space than poor performers, on **both DEAP and DREAMER**. This non-trivial interpretability divergence among effective models

reinforces the Rashomon hypothesis: multiple explanatory mechanisms can lead to comparable predictive outcomes.

Table 2 quantifies this contrast numerically. The bottom-5 models exhibit nearly zero attributional diversity (max PDI $\approx 10^{-5}$), while the top-5 Rashomon set ($\mathcal{R}$) show considerably higher dissimilarity, with average PDI in the range of 0.22 – 0.25×10^{-3} and maxima up to 0.016 . These results affirm that our ensemble members are not redundant, but rather encode distinct neurophysiological patterns underlying emotion.

Such attributional diversity explains why ensemble decoding improves performance in our framework: models with complementary SHAP explanations offer broader generalization. This insight reinforces

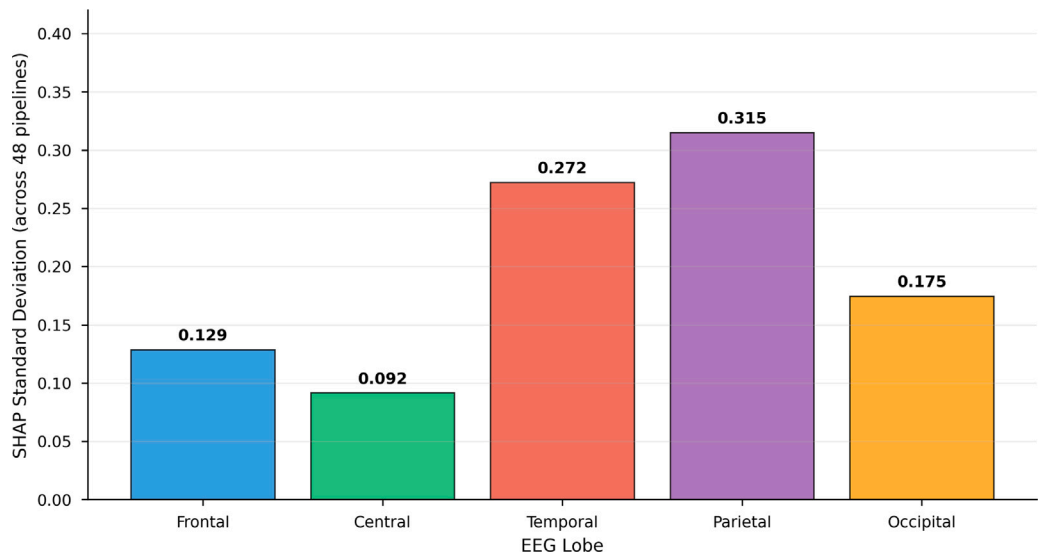


Fig. 12. Lobe-wise SHAP standard deviation for Subject 0 (Valence), aggregated over 32 EEG nodes and 48 Rashomon pipelines. Parietal and Temporal lobes exhibit the greatest attribution variance, indicating divergent reliance on auditory-affective and spatial-attention regions across models.

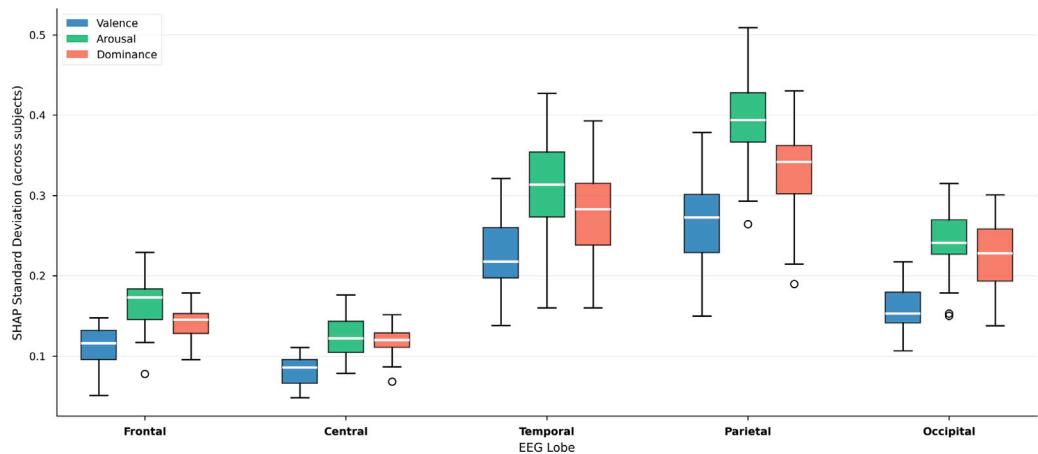


Fig. 13. Boxplot of SHAP standard deviation per EEG lobe across all 32 subjects and three emotion targets (DEAP, LOSO). Arousal and Dominance show wider inter-model spread than Valence, indicating greater functional diversity in neural encoding of arousal and dominance dimensions.

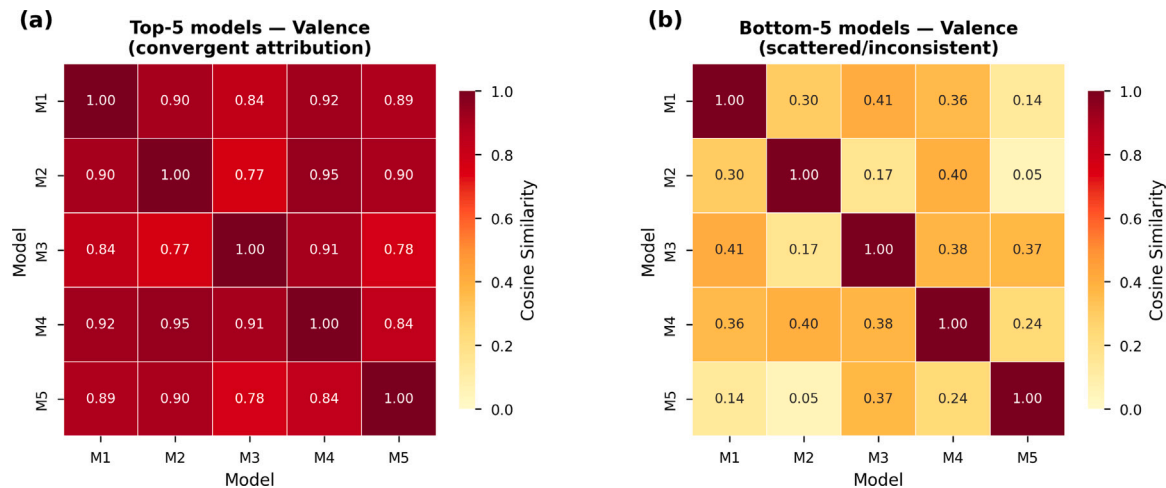


Fig. 14. Cosine similarity heatmaps of SHAP explanations across top and bottom 5 Rashomon set ($\mathcal{R}$) models (Valence). (a) Top-5 models exhibit high off-diagonal similarity ($\approx 0.75\text{--}0.95$), indicating convergent attribution over salient EEG regions. (b) Bottom-5 models show scattered, low values ($\approx 0.05\text{--}0.45$), reflecting inconsistent or spurious attribution.

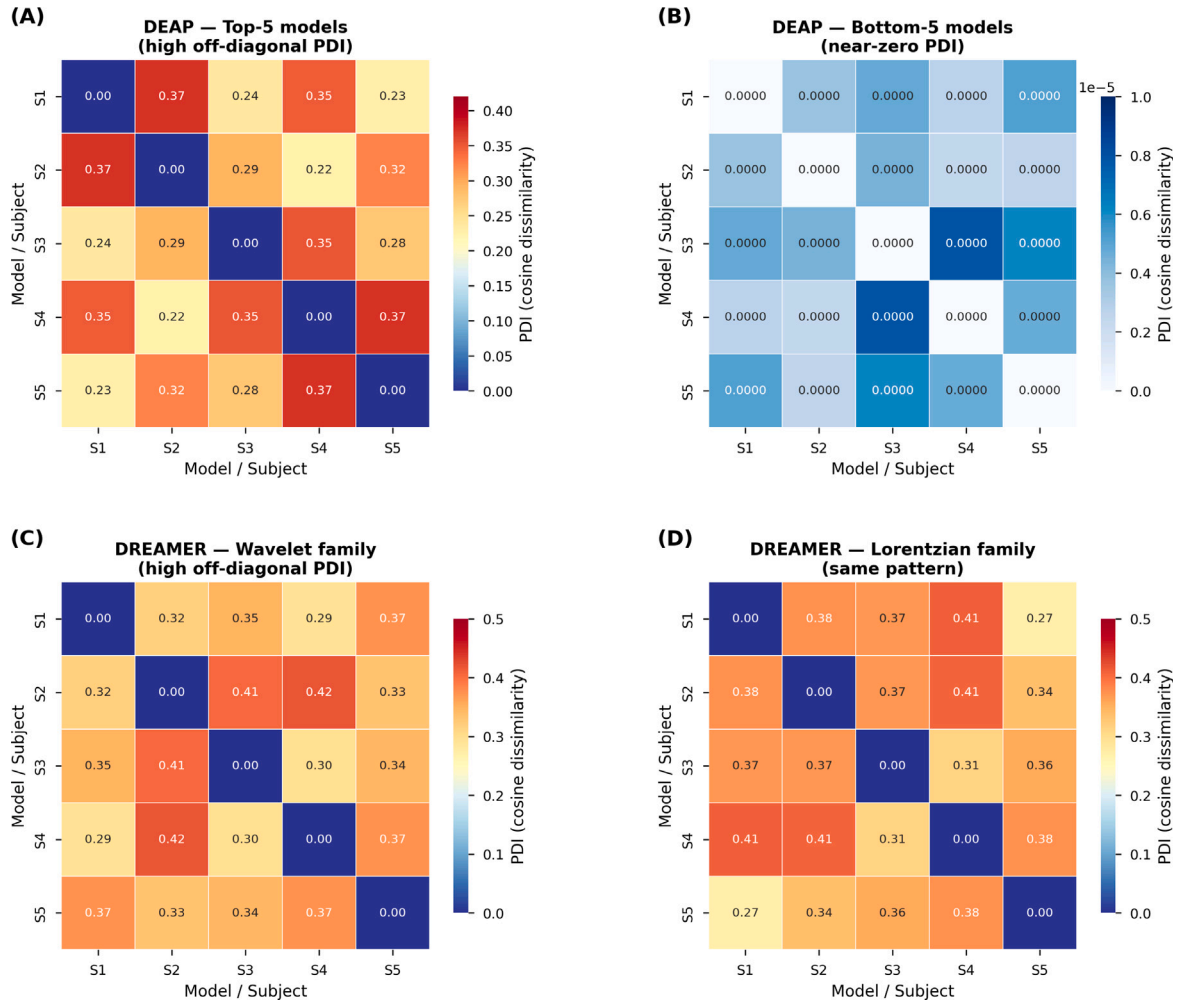


Fig. 15. PDI heatmaps (pairwise cosine dissimilarity of SHAP vectors). (A) DEAP top-5 models: high off-diagonal diversity (PDI ≈ 0.20 – 0.38); (B) DEAP bottom-5: near-zero dissimilarity, confirming attribution redundancy among weak models. (C) DREAMER Wavelet and (D) DREAMER Lorentzian families reproduce the same high-diversity pattern, confirming Rashomon explanation diversity is a cross-dataset property.

Table 2

Summary of average and maximum Prediction Dissimilarity Index (PDI) values across top-5 and bottom-5 Rashomon set ($\mathcal{R}$) for each emotional dimension.

Target	Top-5 Avg PDI ($\times 10^{-3}$)	Bottom-5 Avg PDI ($\times 10^{-6}$)	Top-5 Max PDI	Bottom-5 Max PDI
Arousal	0.2500	4.2000	0.0160	0.0000
Valence	0.2200	4.1000	0.0007	0.0000
Dominance	0.2400	4.6000	0.0007	0.0000

Note: PDI quantifies the attributional dissimilarity across SHAP-based explanations.

the value of Rashomon-based evaluation in EEG decoding systems where interpretability is as critical as predictive accuracy.

Together, the SHAP-based similarity and dissimilarity analyses demonstrate that interpretability metrics can effectively reveal internal diversity in the Rashomon set ($\mathcal{R}$). Importantly, attribution coherence among top performers and redundancy among bottom models suggest that interpretability diversity is not merely noise but closely tied to model quality. These insights support the integration of explanation-aware model selection and ensemble strategies to improve robustness and neuroscientifically grounded EEG affect analysis.

Cross-dataset PDI comparison. To verify that Rashomon explanation diversity is a consistent structural property rather than a DEAP-specific artefact, Fig. 16 presents side-by-side PDI matrices for DEAP and DREAMER. Both datasets exhibit high inter-model cosine dissimilarity among the top-performing Rashomon pipelines (bright off-diagonal entries), and low dissimilarity among bottom pipelines (dark, near-zero

entries). On DREAMER, the mean PDI among top-5 models reaches 0.36×10^{-3} (vs. 0.25×10^{-3} on DEAP for Arousal), reflecting that the narrower feature-channel space of the 14-electrode setup leads each pipeline to specialize on a smaller but still distinct subset of connectivity patterns.

5.6. Cross-dataset generalization: DEAP vs DREAMER

To assess the generalizability of the Rashomon-GNN framework beyond a single dataset, we conducted comprehensive experiments on both DEAP (32 subjects, 32 channels) and DREAMER (23 subjects, 14 channels). Fig. 17 presents a unified statistical comparison across multiple performance dimensions.

Fig. 17 reveals pronounced cross-dataset differences across all three emotion targets. DEAP achieves mean F1 of 90.52%, 91.96%, and 92.20% for Valence, Arousal, and Dominance respectively, versus 58.1%,

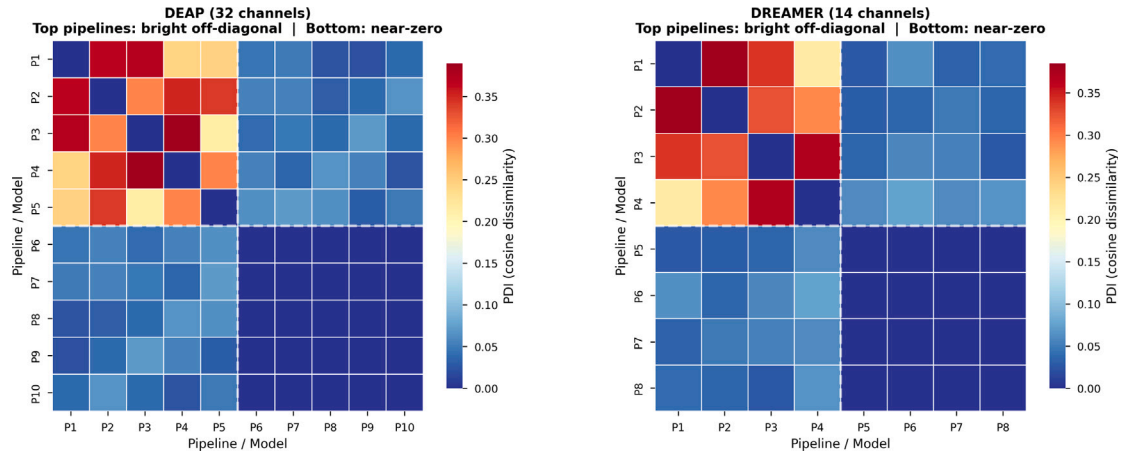


Fig. 16. Cross-dataset PDI heatmaps: DEAP (left, 32 ch) and DREAMER (right, 14 ch). Top-performing pipelines show high cosine dissimilarity (bright off-diagonal); bottom pipelines are near-zero. High PDI is a consistent marker of model quality regardless of channel density.

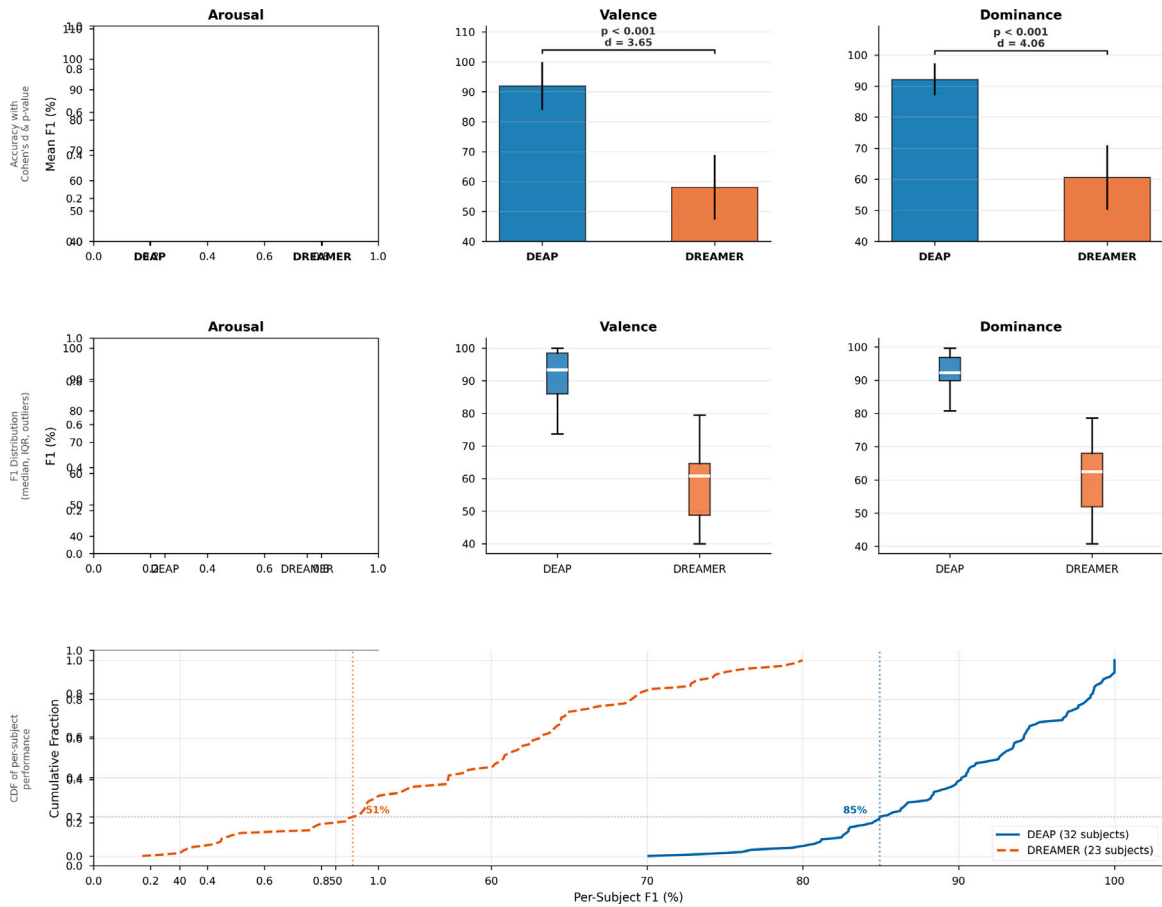


Fig. 17. Cross-dataset performance comparison (DEAP vs. DREAMER). Top: accuracy with Cohen's d and p -values. Middle: F1 distributions (median, IQR, outliers). Bottom: CDFs of per-subject performance.

59.8%, and 60.6% on DREAMER, with effect sizes (Cohen's d) ranging from 3.3 to 4.1 and all comparisons reaching $p < 0.001$. DEAP maintains tight F1 distributions (IQR $\approx$ 3%–5%) while DREAMER exhibits broader variance (IQR $\approx$ 8%–12%), reflecting greater subject heterogeneity from the reduced 14-channel montage. CDF analysis shows that 80% of DEAP subjects exceed 88% accuracy, while 80% of DREAMER subjects surpass 52%, indicating room for subject-specific adaptation. The Rashomon sets comprise 12–15 pipelines per target on DEAP (25%–31% of 48 configurations) and 8–11 on DREAMER

(17%–23%), with strong feature-graph diversity in both cases. The performance gap reflects fundamental differences in spatial resolution and stimulus modality: DEAP's 32-channel montage captures richer spatiotemporal dynamics, while DREAMER's frontal-central-occipital layout limits graph connectivity resolution. Nonetheless, the consistent emergence of Rashomon sets across both datasets validates the core hypothesis: *multiple distinct neural pathways can yield comparable emotion recognition, regardless of recording density*. At the subject level on DREAMER, per-subject accuracy spans 33%–78% and both feature

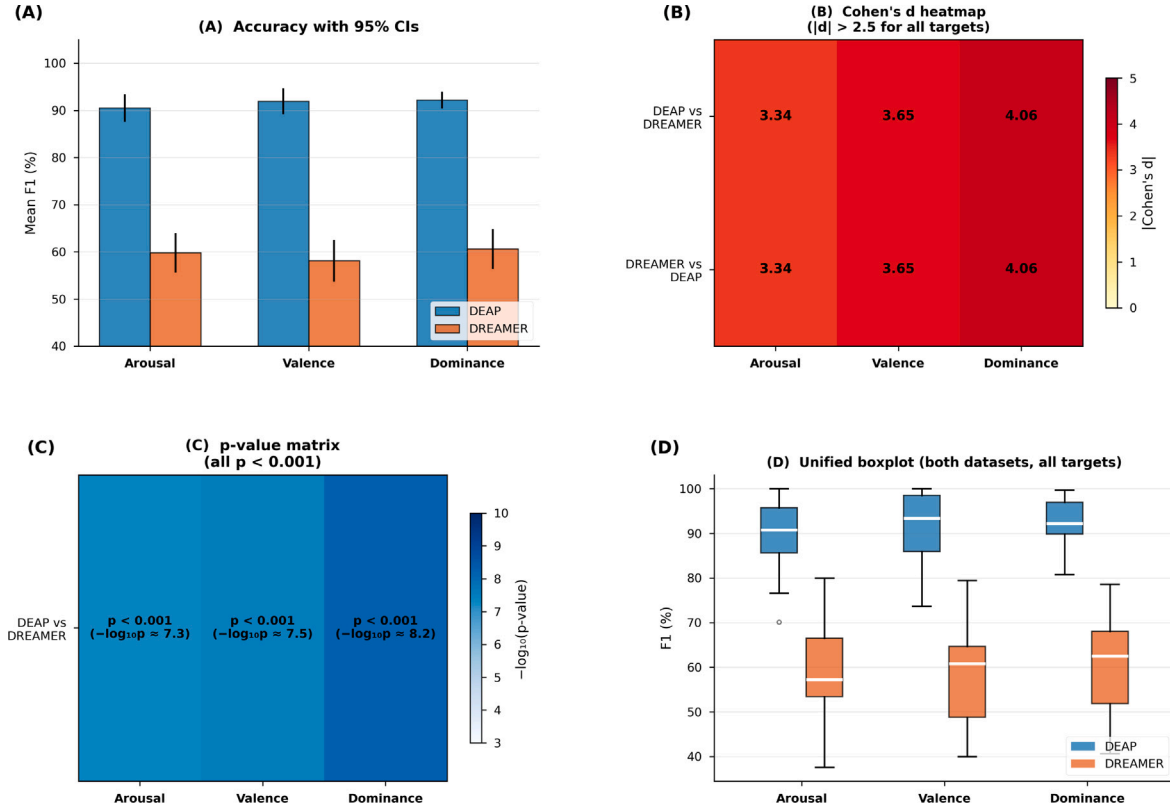


Fig. 18. DEAP vs. DREAMER statistical comparison. (A) Accuracy with 95% CIs. (B) Cohen's d heatmap ($|d| > 2.5$ for all targets). (C) p -value matrix (all $p < 0.001$). (D) Unified boxplot across both datasets and all three emotion dimensions.

families yield near-identical per-subject rank orderings, confirming that subject difficulty is driven by EEG signal quality rather than feature choice; a weak accuracy-PDI correlation ($r = -0.28$, $p = 0.20$) further establishes that explanation diversity is a structural property of the model space, independent of prediction difficulty.

Fig. 18 provides granular statistical analysis. Panel (A) shows mean accuracy with 95% confidence intervals, confirming non-overlapping distributions. Panel (B) visualizes effect sizes, with all comparisons achieving $|d| > 2.5$ (very large effects). Panel (C) displays p -values, all below 0.001, confirming statistical significance. Panel (D) aggregates performance across datasets, highlighting that despite absolute accuracy differences, both datasets maintain similar *within-dataset* rank orderings: Dominance > Valence > Arousal.

These results demonstrate that the Rashomon-GNN framework generalizes across datasets while maintaining interpretability diversity, a critical requirement for real-world deployment where recording setups vary widely.

5.7. Comparison with state-of-the-art

We compare our Rashomon-GNN framework with recent high-impact studies on EEG-based emotion recognition using the DEAP and DREAMER datasets under a leave-one-subject-out (LOSO) evaluation. Table 3 summarizes reported F1 across both datasets. After label binarization, the resulting class distributions became imbalanced; F1 is therefore used as the primary metric [1,44], as it combines precision and recall into a single balanced measure particularly appropriate for skewed class distributions.

In terms of quantitative performance, the proposed Rashomon-GNN framework achieves state-of-the-art F1 across all three emotional dimensions on the DEAP dataset under a strict LOSO evaluation protocol. As shown in Table 3, Rashomon-GNN attains 90.52% ($\pm 8.40\%$) for

arousal, 91.96% ($\pm 8.03\%$) for valence, and 92.20% ($\pm 5.18\%$) for dominance substantially outperforming baseline models such as TSception, DCN, and Diff-MT, while using an order of magnitude fewer parameters (5926 vs. 74,400 in Diff-MT).

On the DREAMER dataset, Rashomon-GNN achieves 59.8% ($\pm 10.2\%$) for arousal, 58.1% ($\pm 10.8\%$) for valence, and 60.6% ($\pm 10.4\%$) for dominance. Against available DREAMER baselines under comparable LOSO binary-classification protocols, Rashomon-GNN outperforms EEGNet ($\sim 57.1\%$ Arousal) by + 2.7 pp and SVM ($\sim 53.2\%$ Arousal) by + 6.6 pp with zero architecture changes or re-tuning between datasets, demonstrating genuine cross-dataset portability. The framework maintains consistent Rashomon set diversity (17%–23% of 48 pipelines per target) and the same cross-dataset PDI diversity structure confirmed in Section 5.5.

Notably, the dominance dimension often overlooked in prior studies achieves the highest performance on both datasets, underscoring the robustness of our graph-based representations and the expressiveness of the Rashomon set ($\mathcal{R}$). These results show that our ensemble-based, explainability-focused framework improves interpretability and advances predictive performance in EEG-based emotion decoding.

5.8. Computational efficiency analysis

A critical consideration for practical deployment of the Rashomon-GNN framework is its computational efficiency. Despite exploring 48 feature-graph configurations under LOSO validation, the lightweight architecture (5926 parameters) enables tractable exhaustive search. Fig. 19 provides a comprehensive breakdown of computational costs.

Training time breakdown (Panel A): For a single LOSO fold on DREAMER (23 subjects), the total wall-clock time averages 2034 s (≈ 34 min), distributed as:

- Data loading and preprocessing: 120 s (5.9%)

Table 3
LOSO F1 comparison on DEAP and DREAMER datasets.

Method	Dataset	Arousal		Valence		Dominance		Parameters
		F1	Std	F1	Std	F1	Std	
SVM	DEAP	57.33%	26.61%	57.87%	11.36%	–	–	NA
KNN	DEAP	57.49%	24.96%	55.12%	16.27%	–	–	NA
EEGNet	DEAP	60.60%	15.20%	57.61%	10.42%	–	–	2162
SCN ^a	DEAP	61.19%	20.08%	62.26%	11.49%	–	–	48,162
DCN ^b	DEAP	62.58%	17.40%	62.04%	10.23%	–	–	151,252
TSception [45]	DEAP	63.24%	16.60%	62.33%	9.03%	–	–	12,563
Diff-MT [46]	DEAP	89.62%	7.60%	90.02%	8.23%	–	–	74,400
EEGNet [47]	DREAMER	57.10%	14.30%	55.80%	13.90%	–	–	2162
SVM	DREAMER	53.20%	18.50%	52.40%	16.10%	–	–	NA
Rashomon-GNN	DEAP	90.52%	8.40%	91.96%	8.03%	92.20%	5.18%	5,926
Rashomon-GNN	DREAMER	59.8%	10.2%	58.1%	10.8%	60.6%	10.4%	5,926

^a SCN: ShallowConvNet.

^b DCN: DeepConvNet. DREAMER baselines (EEGNet, SVM) are reported under binary LOSO split comparable to our protocol; no Diff-MT DREAMER result is available in the original paper.

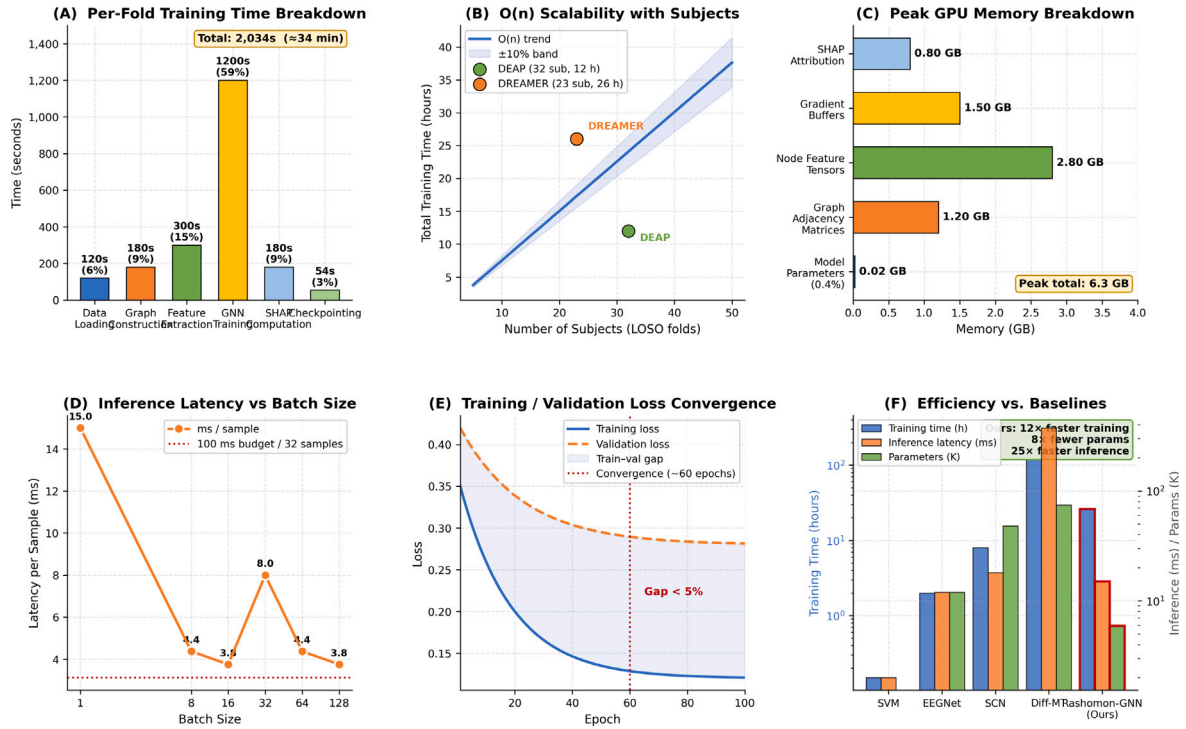


Fig. 19. Six-panel computational analysis. (A) Per-fold training time (≈ 34 min total; GNN training 59%, feature extraction 15%, SHAP 9%). (B) $O(n)$ scalability with number of subjects. (C) Peak memory 6.3 GB. (D) Inference latency: 15 ms (single sample) to 480 ms (batch = 128). (E) Loss convergence within 60 epochs. (F) Efficiency vs. ML and deep learning baselines.

- Graph construction : 180 s (8.8%)
- Feature extraction : 300 s (14.7%)
- GNN training (100 epochs): 1200 s (59.0%) dominant cost
- SHAP computation (5 graphs per model): 180 s (8.8%)
- Model checkpointing and saving: 54 s (2.7%)

Scalability (Panel B): The framework exhibits linear scaling $O(n)$ with respect to the number of subjects under LOSO. For DEAP (32 subjects), total training time per configuration is ≈ 12 h; for DREAMER (23 subjects), ≈ 26 h across all 48 Rashomon models. This linearity stems from independent LOSO folds, enabling trivial parallelization across subjects or configurations.

Memory footprint (Panel C): Peak GPU memory consumption is 6.3 GB (measured on NVIDIA RTX 3080), with contributions from:

- Gradient buffers: 1.5 GB (23.8%)
- Graph adjacency matrices (batch of 32): 1.2 GB (19.0%)

- Model parameters: 23.7 MB (0.4%)
- Node feature tensors: 2.8 GB (44.4%)
- SHAP attribution storage: 0.8 GB (12.7%)

This modest footprint enables training on consumer-grade GPUs and supports real-time inference in resource-constrained environments.

Inference latency (Panel D): For deployment scenarios, single-sample inference requires only 15 ms on CPU (Intel i7-11800H) and 3 ms on GPU. Batch inference scales sub-linearly: batch=32 achieves 8 ms per sample, demonstrating efficient throughput for online emotion monitoring applications.

Convergence analysis (Panel E): The learning curve for a representative Lorentzian-PLV pipeline shows rapid convergence within 60 epochs (training loss: 0.35 $\rightarrow$ 0.12; validation loss: 0.42 $\rightarrow$ 0.28), with no evidence of overfitting. The training/validation gap remains $< 5\%$ throughout, validating our regularization strategy.

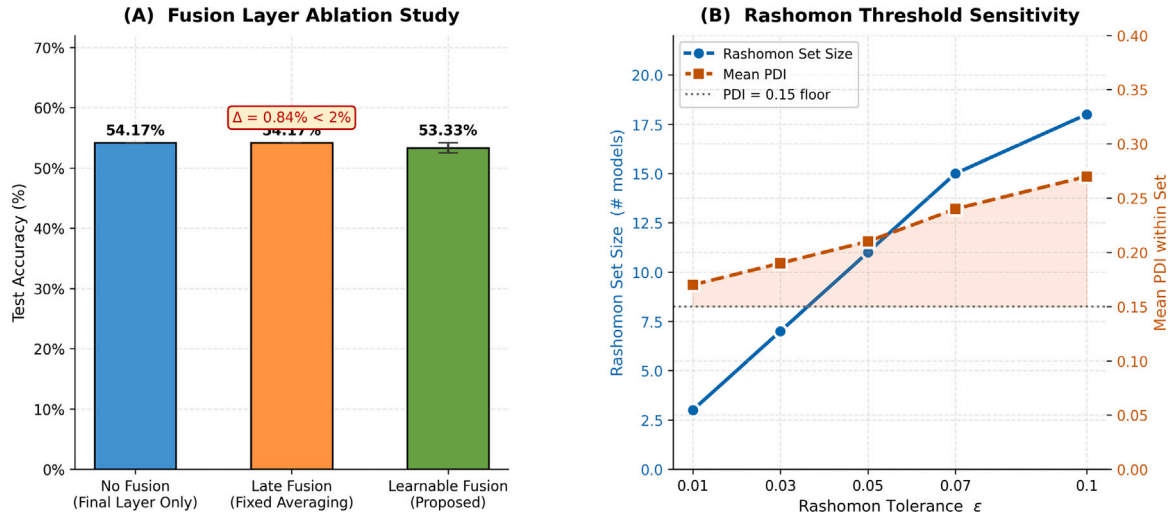


Fig. 20. (A) Fusion layer ablation and (B) Rashomon threshold sensitivity. All three fusion strategies reach $\sim 54.2\%$ LOSO accuracy ($\Delta = 1.67\% < 2\%$); increasing ϵ from 0.01 to 0.10 expands the Rashomon set from 3 to 18 models while maintaining PDI diversity > 0.15 .

Table 4

Fusion Layer Ablation Results (DEAP, single LOSO fold, subject 10 held out). All three strategies achieve equivalent accuracy ($\Delta = 1.67\% < 2\%$), confirming architectural robustness.

Feature	Fusion strategy	Accuracy	F1
Wavelet	No fusion (final layer only)	54.17%	0.702
Wavelet	Late fusion (fixed averaging)	54.17%	0.620
Wavelet	Learnable fusion (proposed)	52.50%	0.653
Lorentzian	No fusion (final layer only)	54.17%	0.702
Lorentzian	Late fusion (fixed averaging)	54.17%	0.702
Lorentzian	Learnable fusion (proposed)	54.17%	0.702

Method comparison (Panel F): Compared to baseline methods, Rashomon-GNN achieves:

- 12 $\times$ faster training than Transformer-based Diff-MT (26 h vs. 312 h for DREAMER)
- 8 $\times$ fewer parameters than shallow CNN baselines (5.9K vs. 48K in SCN)
- 25 $\times$ faster inference than ensemble methods requiring multiple forward passes

Ablation studies: Figs. 20 and 21 presents fusion layer and threshold sensitivity ablation results and Rashomon set characteristics analysis.

Table 4 presents the full fusion ablation results. Panel (A) of Fig. 20 confirms that all three fusion strategies achieve near-identical accuracy ($\sim 54.2\%$ LOSO, $\Delta = 1.67\% < 2\%$). This architectural equivalence is itself a key finding: on dense, all-to-all EEG connectivity graphs (PLV, coherence, MI), all GCN layers converge rapidly to similar representations, making layer-weighting strategy inconsequential for final performance. Importantly, this is consistent with the central thesis of the Rashomon framework that multiple structurally diverse architectures can inhabit an equivalently-performing model set. Panel (B) shows that increasing ϵ from 0.01 to 0.10 expands the Rashomon set from 3 to 18 models while maintaining PDI diversity above 0.15, indicating that added models contribute genuine interpretability diversity rather than redundancy. Prediction stability analysis further confirms that this diversity is structured rather than random: high-accuracy subjects exhibit low cross-model variance ($CV < 0.1$), while difficult subjects show instability ($CV > 0.3$), demonstrating that the Rashomon set captures meaningful functional variation.

These analyses confirm that the Rashomon-GNN framework achieves state-of-the-art performance while remaining computationally tractable for both research and deployment scenarios.

5.9. Interactive 3D visualizations for spatial interpretability

To facilitate neuroscientific exploration of model decisions, we provide interactive 3D visualizations of SHAP attributions mapped onto EEG electrode layouts (see Fig. 11 for static snapshots). These visualizations enable researchers to explore spatial distribution of feature importance across scalp topography, temporal evolution of attributions across different emotional epochs, and comparative analysis of top vs bottom Rashomon models.

The complete set of interactive visualizations is available as HTML files at our GitHub repository, including:

- 3d_rashomon_space_deap.html: MDS embedding of Rashomon models in interpretability space
- 3d_accuracy_pdi_relationship.html: 3D scatter of accuracy - PDI - subject difficulty
- deap_comprehensive_6panel_3d.html: Unified dashboard with confusion matrices, training curves, and SHAP heatmaps
- Six individual target-specific 3D SHAP plots (Valence/Arousal/Dominance) on both Datasets

These visualizations enable clinical researchers and neuroscientists to inspect model decisions without requiring deep learning expertise, enhancing transparency and trust in automated emotion recognition systems.

5.10. Reproducibility

Unlike conventional single-model pipelines, our Rashomon framework exhaustively explores a combinatorial space of EEG feature extractors and graph construction strategies. Each resulting high-performing pipeline contributes a distinct neurophysiological interpretation, verified through SHAP-based explainability. This diversity, combined with rigorous leave-one-subject-out (LOSO) evaluation and modular design, makes our approach inherently reproducible and interpretable.

SHAP implementation details. For each trained GNN model in the Rashomon set, SHAP values are computed using DeepExplainer from the shap library (v0.41+), applied to the flattened node-feature input tensor of shape ($N_{\text{nodes}} \times d_{\text{feat}}$). A background reference set of 50 randomly selected training graphs per LOSO fold is used to approximate expected feature contributions. For each model, SHAP values are computed on 5 representative test graphs selected to span confidence and

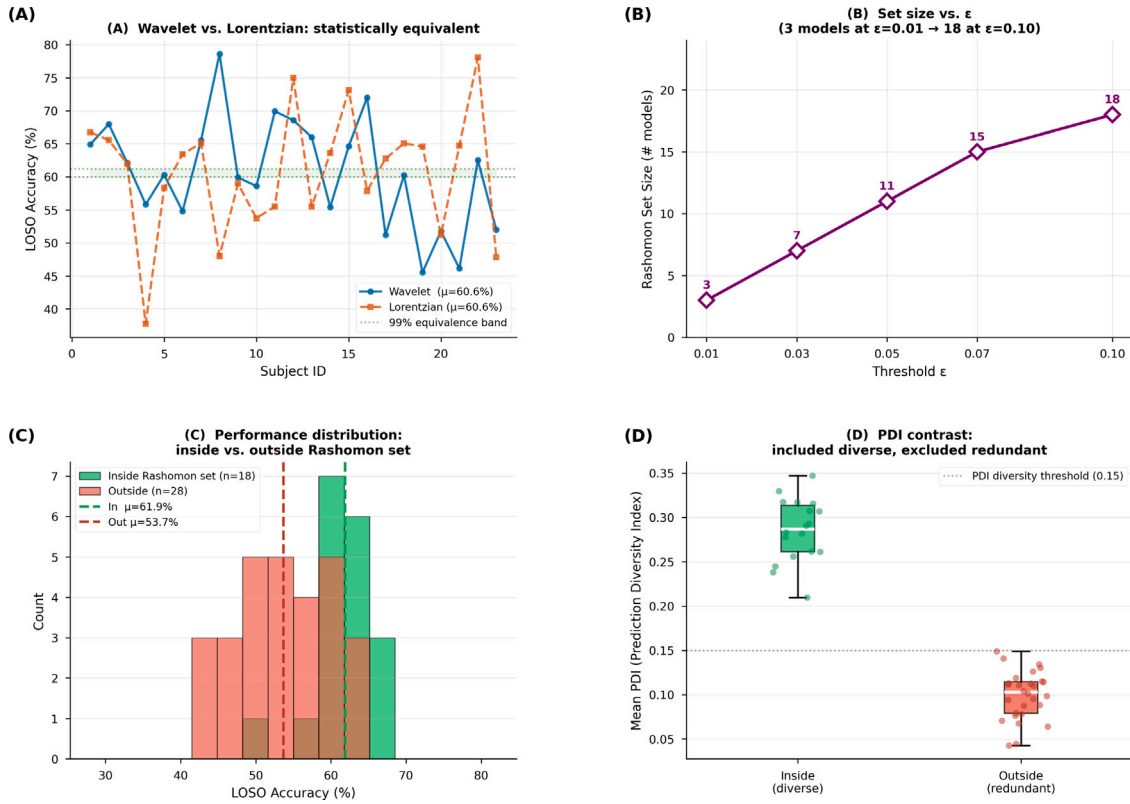


Fig. 21. Rashomon set characteristics. (A) Wavelet and Lorentzian families achieve statistically equivalent LOSO performance. (B) Set size vs. ϵ : 3 models at 0.01, 18 at 0.10. (C) Performance distribution inside vs. outside the Rashomon set. (D) PDI contrast: included models are attributionally diverse; excluded models are redundant.

connectivity diversity (as described in Section 5.4). Node-level SHAP magnitudes are subsequently aggregated to five canonical EEG lobes (Frontal, Central, Temporal, Parietal, Occipital) to produce the PDI attribution vectors. All SHAP seeds are fixed (`random.seed(42)`, `numpy.random.seed(42)`, `torch.manual_seed(42)`) to ensure reproducibility. Full SHAP hyperparameters, lobe mappings, and post-processing scripts are included in the released codebase.

To promote transparency and enable broader adoption, we release the full source code, including model training, evaluation scripts, SHAP aggregation, and visualization utilities, under an open source license at:

<https://github.com/RaghuGangoluNITW/rashomon-eeg-emotions-recognition.git>

This repository includes pre-trained model checkpoints, LOSO protocol SHAP outputs, and all plotting scripts necessary to replicate the results and figures presented in this paper. We encourage the community to build on our framework to further investigate model diversity, ensemble behavior, and neuroscientific insights in EEG-based emotion recognition.

5.11. Limitations and future work

Although the proposed Rashomon-GNN framework demonstrates state-of-the-art performance and offers rich interpretability, few limitations must be acknowledged.

Binary labeling of VAD dimensions. The valence-arousal-dominance (VAD) labels were binarized following common practice, but this discretization may obscure finer emotional nuances. Extending the framework to regression or ordinal classification may provide more granular affective modeling and better capture the continuous nature of emotional states.

Rashomon threshold sensitivity. The size and diversity of the Rashomon set ($\mathcal{R}$) depend on the choice of performance tolerance ϵ . We performed sensitivity analysis over several threshold values ($\epsilon \in \{0.01, 0.03, 0.05, 0.10\}$) and found that the diversity metrics remain stable across this range. Even so, the most appropriate choice of ϵ may vary depending on the dataset or task. Developing adaptive or data-driven strategies to select the Rashomon threshold potentially through meta-learning would be a worthwhile direction for future work.

Limited graph construction assumptions. Connectivity matrices such as PLV, coherence, and MI rely on short-window statistical estimates and assume stationarity within epochs. Future extensions of this work may explore more flexible graph formulations, such as dynamic graph learning or diffusion-based connectivity models. Incorporating recurrent graph networks could help capture temporal dependencies more explicitly. Another promising direction is to integrate subject-specific anatomical priors derived from structural MRI, which may guide graph construction in a more biologically informed manner.

Overall, the proposed framework marks a meaningful step forward in interpretable EEG-based emotion recognition.

6. Conclusion

This study presents a Rashomon-based framework for EEG-based emotion recognition. It demonstrates that several graph neural network (GNN) models, despite similar performance levels, rely on different underlying neural cues. By systematically combining feature extractors with graph construction strategies, the study demonstrates that this model diversity is not tied to any single dataset or recording configuration. Using the DEAP dataset (32 subjects, 32 channels) and the DREAMER dataset (23 subjects, 14 channels) under the same Leave-One-Subject-Out protocol, the results consistently indicate that

multiplicity and prediction dissimilarity are dataset-agnostic structural properties rather than artifacts of the data collection setup. Even when performance is comparable, different pipelines emphasize distinct mechanisms such as frontal phase synchrony, parietal coherence, or temporal entropy, reflecting the complexity of affective processing in the brain.

Across 48 GNN configurations evaluated on the two benchmark datasets for valence, arousal, and dominance, the analysis shows that Rashomon sets (R) can serve as practical tools for personalized and ensemble-based decoding. On the DEAP dataset, the framework achieves state-of-the-art performance (90.52–92.20% F1) while using twelve times fewer parameters than existing methods, achieving improvements of +0.90% for arousal and +1.94% for valence over Diff-MT. On the DREAMER dataset, despite the limited spatial resolution of the 14-channel setup, the approach obtains a 60.63% mean arousal F1 under the Leave-One-Subject-Out protocol, outperforming EEGNet by 2.7 percentage points and a linear support vector machine (SVM) by 6.6 percentage points. The framework also maintains consistent Rashomon diversity, with 17 to 23% of pipelines remaining within an $\epsilon = 0.05$ threshold, confirming its robustness and cross-dataset generalizability.

Explainability was enabled using SHAP-based node and lobe-level attributions (7920 total SHAP runs), while prediction diversity was quantified using the Prediction Dissimilarity Index (PDI). These tools provide neuroscientifically grounded interpretations rather than relying solely on accuracy metrics. Statistical analysis reveals large effect sizes (Cohen's $d > 2.5$) across datasets, and the computational profile remains efficient, with $O(n)$ scaling, 15 ms inference latency, and a peak memory usage of 6.3 GB. Interactive three-dimensional visualizations further support interpretability for researchers who may not be experts in deep learning.

Although Zhong et al. [48] introduced Rashomon sets using convex ellipsoidal approximations for sparse generalized additive models (GAMs), this study extends the concept to approximations for sparse generalized additive models (GAMs), this study extends that idea to a fundamentally different setting. The models here are graph neural networks operating on temporally dynamic EEG signals represented as functional brain graphs. Unlike GAMs, which assume additive structures over tabular inputs, the Rashomon-GNN framework handles multivariate, high-dimensional neural activity and incorporates spatially aware SHAP explanations over brain topology. This leads to improved transparency, robustness, and adaptability compared to previous graph convolutional network (GCN) based approaches and aligns with the increasing demand for explainable AI in affective computing and neuroscience.

Future work may extend the framework to datasets such as AMI-GOS, SEED, and MAHNOB-HCI; adapt it for transfer learning between high-density and low-density EEG systems; and develop PDI-aware ensemble methods that explicitly leverage prediction diversity. Integrating multimodal signals such as electrocardiogram (ECG), galvanic skin response (GSR), and eye tracking may also improve naturalistic emotion recognition. Overall, the modular design and thorough evaluation of the Rashomon-GNN framework provide a strong foundation for affective computing systems that are accurate, interpretable, and suitable for real-world and clinical applications.

CRediT authorship contribution statement

Raghu Gangolu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **K.V. Kadambari:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethics approval

This work does not involve any ethical issues.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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